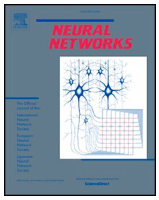
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Full Length Article



High-order neural network extended state-observer with nuclear application

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ABSTRACT

A high-order neural network extended state observer (HoNN-ESO) is proposed for single-output nonlinear disturbed systems. Under specified condition within a defined set, the HoNN-ESO delivers globally asymptotically bounded observations for state-variables, lumped disturbance and its high-order time-derivatives. By synergistically coupling a multi-layer perception (MLP) with an extended state-observer (ESO), the HoNN-ESO effecitively eliminates the need for high observer gains to compensate for system uncertainties. The inherent unknown nonlinearities in system dynamics are approximated by a custom-designed single-hidden-layer MLP equipped with an online learning law. Increasing the number of hidden-layer neurons can further enhance the overall observation performance. The HoNN-ESO is subsequently applied to the state and disturbance observation of a modular high temperature gas-cooled reactor. Simulation results across five representative scenarios not only validate the theoretical derivations and demonstrate the feasibility of HoNN-ESO in generating smart indicators for anomaly detection, but also verify that satisfactory observation performance can be reliably achieved by appropriately increasing the neuron count.

1. Introduction

The rapid development and large-scale deployment of Industrial Internet have significantly improved the networking capacity of critical infrastructure assets such as power grids, district heating networks (DHNs), and nuclear power plants (NPPs). Accordingly, intelligent anomaly detection and dependable control strategies are in urgent demand. Within this context, state observation occupies a pivotal position: it reconstructs immeasurable internal states of dynamic systems, generates characteristic indicators for anomaly detection, and supports state-feedback controller design to improve control performance (Luenberger, 1964). Thanks to their powerful function-approximation capability, neural networks (NNs) have been widely deployed for the identification and control of nonlinear systems (Narendra & Parthasarathy, 1990; Wang & Li, 2022; Zhang & Wang, 2022), with application covering robot manipulators (Rubio et al., 2024) and boiler-turbine sets (Li et al., 2025). In particular, neural network observers (NNOs), which integrates neural networks into state-observer frameworks, have become a major research topic in system monitoring and control, yielding numerous meaningful results over recent decades.

Since 1990s, various recurrent neural networks (RNNs) have been adopted for the identification and control of nonlinear systems under the assumption that the full state-vectors are fully measurable (Kosmatopoulos et al., 1995; Merabet et al., 2023; Poznyak et al., 1999; Sanchez & Ricalde, 2003). Meanwhile, advances in nonlinear control and observation have spurred the development of sophisticated neural architectures, including delayed-competitive neural networks (Cao et al., 2024) and memristive inertial competitive neural networks (Subhashri et al., 2026). RNNs have also been adapted for state observation tasks, where Jacobian matrices are commonly used as the state-matrix of NNOs and feedback signals are constructed from output observation errors. A representative work is the linearly parameterized recurrent high order neural observer (RHONO) presented in (Ricalde & Sanchez, 2005), where the network connections are adaptively updated online using an inverse optimal learning law. This RHONO achieves asymptotically convergent state-observation provided that the Kalman-Yakuboich-Popov (KYP) condition is satisfied (Rantzer, 1996). The RHONO framework was later extended to discrete-time nonlinear systems (Alanis et al., 2011), where the extended Kalman filter (EKF)

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(Haykin, 2021) serves as an online training mechanism to guarantee uniformly ultimately bounded (UUB) observation errors. Reference (Kim et al., 1996) gives a linearly parameterized MLP-based NNO that also enforces the KYP condition. Based on the KYP-Lyapunov synthesis framework established in (Kim et al., 1996; Ricalde & Sanchez, 2005), a robust adaptive NNO has been proposed to preserve UUB observation performance in the presence of bounded external disturbances (Zhu et al., 1997). A radial basis function (RBF)-driven NNO has also been reported, wherein a projection-operator based learning law is derived following the same KYP-Lyapunov methodology (Stepanyan & Hovakimyan, 2007). It can be concluded that the KYP condition acts as a fundamental prerequisite for deriving NNO learning laws via Lyapunov's direct method, which imposes stringent limitations on system output structures.

Circumventing constraints imposed by the KYP condition remains a central challenge for NNO development. Existing literature (Vargas & Hemerly, 2001) reveals that the KYP condition can be discarded if the system state-matrix is diagonally Hurwitz; an adaptive linearly-parameterized NNO is then constructed to yield bounded observation. Nevertheless, practical engineering systems rarely possess diagonally Hurwitz state matrices. Subsequent investigations (Chairez, 2009; Chairez et al., 2006; Chairez et al., 2009) indicate that the KYP condition can be bypassed for zero-state detectable systems, giving rise to several learning-law solutions for linearly-parameterized NNOs. Since the KYP condition originates from the application of Lyapunov's direct method, alternative training strategies may help avoid this requirement. Motivated by this insight, reference (Abdollahi et al., 2006) employs a gradient learning mechanism to design a nonlinearly-parameterized MLP-based NNO, at the cost of requiring non-singularity of system matrices. Similarly, a two-hidden-layer MLP-based NNO has been developed for fault detection and identification within satellite attitude control systems (Talebi et al., 2009). Apart from gradient-based approaches, deterministic-learning (DL)-based NNOs have attracted considerable attention (Dinh et al., 2014; Hu et al., 2022; Wang & Hill, 2010). By combining a high gain observer (HGO) with a dynamic RBF identifier, a DL-based NNO is given for single-input-single-output (SISO) nonlinear systems with recurrent trajectories (Wang & Hill, 2010). Asymptotic observation is guaranteed by the persistent excitation (PE) condition rather than the KYP condition. This DL framework has been further extended to discrete-time nonlinear systems with recurrent trajectories (Hu et al., 2022). For multi-input-multi-output (MIMO) nonlinear systems, an MLP-based NNO with a single hidden layer is proposed in (Dinh et al., 2014). The KYP condition is eliminated by incorporating a sliding-mode dynamic filter (Xian et al., 2004), on the condition that all high-order derivatives of system output are measurable. Driven by the advances in artificial intelligence, deep-learning networks (LeCun et al., 2015) and the long short-term memory (LSTM) networks (Hochreiter & Schmidhuber, 1997) are integrated with the dynamic-filtering approach in (Dinh et al., 2014) for NNO design. This yields both multi-hidden-layer MLP based NNOs (Dixon, 2023) and LSTM-based NNOs specifically for second-order nonlinear systems (Griffis et al., 2024). The learning laws for deep MLPs in (Dixon, 2023) and LSTM networks in (Griffis et al., 2024) are derived via Lyapunov's direct method, accounting for first-order neural-network approximation error and driven by error signals generated from associated dynamic filters. Very recently, neural network extended state observers (NNESO) have emerged, where the KYP condition is replaced by a matching condition for system nonlinearities (Zhu et al., 2023). Although NNESO achieves asymptotically bounded observations of lumped disturbances and their first-order time-derivative, its practical applicability is restricted by the prerequisite that full state-vectors must be measurable.

In this paper, a high-order neural network extended state observer (HoNN-ESO) is proposed for disturbed autonomous single-output nonlinear systems. A single-hidden-layer MLP is embedded into the canonical ESO structure so that both KYP and PE conditions can be avoided. The HoNN-ESO synergistically combines the disturbance-observation strengths of ESO and the nonlinearity-identification capability of MLP, as conceptually illustrated in Fig. 1. Notably, the embedded MLP does not merely perform static compensation for ESO outputs. Instead, MLP weights are augmented as additional extended states to facilitate online identification. Specifically, the extended state vector of HoNN-ESO contains both ESO internal states and MLP weight estimates, which are updated by an online learning law derived from Lyapunov's direct method. Theoretical analysis proves that the proposed HoNN-ESO equipped with this Lyapunov-based learning law achieves globally bounded observation and identification, making it well-suited for generating smart detecting variables for system monitoring and anomaly detection. The HoNN-ESO is applied to state and disturbance observation for a high temperature gas-cooled reactor, and numerical simulation validates its feasibility and satisfactory performance.

Table 1 summarizes comprehensive comparisons between the proposed HoNN-ESO and existing NNOs/ESOs reviewed above. Unlike shallow NNOs reported in (Abdollahi et al., 2006; Kim et al., 1996; Ricalde & Sanchez, 2005; Stepanyan & Hovakimyan, 2007; Vargas & Hemerly, 2001; Zhu et al., 1997), which only observe internal states using linear shallow neural networks, HoNN-ESO reconstructs full state-vectors as well as disturbances and their high-order time--derivatives while removing the KYP requirement through a nonlinear-matching condition. By contrast to DL-based NNOs in (Dinh et al., 2014; Hu et al., 2022) targeting systems with recurrent trajectories, HoNN-ESO works for systems with arbitrarily bounded trajectories without enforcing the PE condition. Compared with conventional ESO (Guo & Zhao, 2011; Han, 2009) which focuses purely on disturbance observation, HoNN-ESO additionally identifies intrinsic nonlinearities embedded within state and disturbance dynamics. Sliding mode NNO (Xian et al., 2004), NN-ESO (Zhu et al., 2023) and multi-hidden layer NNO (Dixon, 2023) require either high-order output derivatives or complete state-measurements. HoNN-ESO overcome such limitations and simultaneously observes unmeasurable internal states and identifies dynamic nonlinearities. Though LSTM-based NNO (Griffis et al., 2024) can realize state observation and nonlinearity identification, it is confied to second-order systems, whereas HoNN-ESO is applicable to systems of arbitrary orders. These comparisons demonstrates the prominent advantages of HoNN-ESO for state observation and nonlinearity identification of high-order disturbed nonlinear dynamical systems.

The rest of this paper is structured as follows. Section 2 formulates the studied single-output disturbed nonlinear system elaborates on the HoNN-ESO design for state and disturbance observation. Section 3 derives the online learning law and establishes conditions for globally asymptotically bounded properties. Section 4 implements HoNN-ESO for nuclear reactor monitoring and presents numerical simulation results and detailed discussions. Finally, Section 5 draws conclusions and outlines directions for future research.

2. High-order neural network extended-state observer

Consider the single-output disturbed autonomous nonlinear systems:

$$\begin{cases} \dot{x}_i = x_{i+1} + f_i(\mathbf{x}), \quad i = 1, \cdots, n-1, \\ \dot{x}_n = f_n(\mathbf{x}) + \xi, \\ y = x_1, \end{cases} \tag{1}$$

where $\boldsymbol{x}=[x_1, \cdots, x_n]^{\mathrm{T}} \in \mathrm{R}^n$ denotes the state-vector, $y \in \mathrm{R}$ is the system output, f_i: $\mathrm{R}^{n+1} \to \mathrm{R}$ represent norm-boundedly smooth functions ($i = 1, \ldots, n$), $\xi \in \mathrm{R}$ is external disturbance. Without loss of generality, the dynamics of disturbance ξ is expressed as:

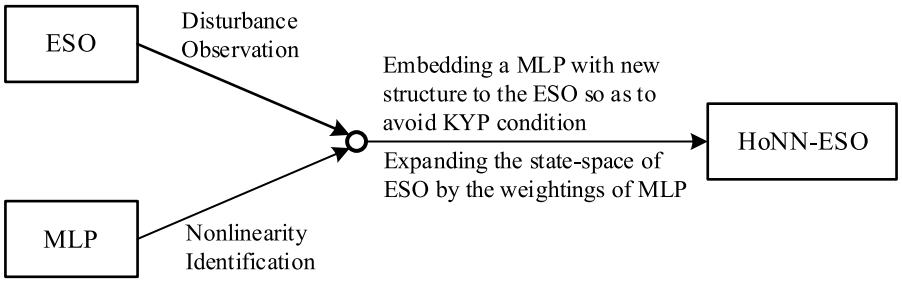


Fig. 1. Graphical idea, ESO: extended state-observer, MLP: multi-layer perception, HoNN-ESO: high-order neural network extended state-observer, KYP: Kalman-Yakuboich-Popov.

Table 1
Comparison among the cited and proposed neural network observers.

State-Observers	Identifying nonlinear dynamics	Observing disturbances	Satisfying the KYP condition	Satisfying the PE condition	Needing to measure the time-derivatives of output	Needing to measure all the states	System Taking recurrent trajectories	System order is NOT larger than two
Shallow NNO (Abdollahi et al., 2006; Kim et al., 1996; Ricalde & Sanchez, 2005; Stepanyan & Hovakimyan, 2007; Vargas & Hemerly, 2001; Zhu et al., 1997)	Y	N	Y	N	N	N	N	N
DL-based NNO (Dinh et al., 2014; Hu et al., 2022)	Y	N	N	Y	N	N	Y	N
Sliding mode NNO (Xian et al., 2004)	Y	N	N	N	Y	N	N	N
Multi-hidden layer NNO (Dixon, 2023)	Y	N	N	N	N	Y	N	N
LSTM NNO (Griffis et al., 2024)	Y	N	N	N	N	N	N	Y
NNESO (Zhu et al., 2023)	Y	Y	N	N	N	Y	N	N
ESO (Guo & Zhao, 2011; Han, 2009)	N	Y	N	N	N	N	N	N
Proposed HoNN-ESO	Y	Y	N	N	N	N	N	N

NNO: neural network observer, DL: deterministic learning, LSTM: long short-term memory, ESO: extended state-observer, NNESO: neural network extended-state observer, HoNN-ESO: high order neural network extended observer, Y: Yes, N: No.

$$\begin{cases} \dot{\zeta}_j = f_{\mathrm{d},j}(\boldsymbol{\zeta}) + \zeta_{j+1}, \quad j = 1, \cdots, m-1, \\ \dot{\zeta}_m = f_{\mathrm{d},m}(\boldsymbol{\zeta}) + d, \\ \xi = \zeta_1, \end{cases} \tag{2}$$

where $\boldsymbol{\zeta}=[\zeta_1, \cdots, \zeta_m]^{\mathrm{T}}\in\mathrm{R}^m$ is the state-vector of disturbance dynamics, $f_{\mathrm{d},i}$: $\mathrm{R}^{n+1}\to\mathrm{R}$ stand for norm-bounded smooth nonlinear terms associated with disturbance evolution, and scalar $d\in\mathrm{R}$ satisfies bounded constraint

$$|d| \le D, \tag{3}$$

with D as a positive bounded constant.

Treat ζ_i ($i=1,\ldots, m$) given by (2) as extended state variables, and define the extended state-vector as

$$\mathbf{z} = \begin{bmatrix} \mathbf{x} \\ \boldsymbol{\zeta} \end{bmatrix} \in \mathrm{R}^N, \tag{4}$$

with total dimension $N=n+m$. The extended state-space model is obtained by combining original system dynamics (1) and disturbance dynamics (2):

$$\begin{cases} \dot{\mathbf{z}} = \mathbf{A}_{\mathrm{o}}\mathbf{z} + \mathbf{f}(\mathbf{z}) + \mathbf{d}, \\ \mathbf{y} = \mathbf{c}\mathbf{z}, \end{cases} \tag{5}$$

where

$$\mathbf{A}_{\mathrm{o}} = \begin{bmatrix} \mathbf{O}_{(N-1)\times 1} & \mathbf{I}_{N-1} \\ 0 & \mathbf{O}_{1\times(N-1)} \end{bmatrix}, \tag{6}$$

$$\mathbf{f}(\mathbf{z}, u) = \begin{bmatrix} f_1(\mathbf{x}, u) & \cdots & f_n(\mathbf{x}, u) & f_{\mathrm{d},1}(\boldsymbol{\zeta}) & \cdots & f_{\mathrm{d},n}(\boldsymbol{\zeta}) \end{bmatrix}^{\mathrm{T}}, \tag{7}$$

$$\mathbf{d} = \begin{bmatrix} \mathbf{O}_{1\times(N-1)} & d \end{bmatrix}^{\mathrm{T}}, \tag{8}$$

and

$$\mathbf{c} = \begin{bmatrix} 1 & \mathbf{O}_{1\times(N-1)} \end{bmatrix}. \tag{9}$$

Assume that a single-hidden-layer multilayer perception (MLP) Φ satisfies

$$\Phi(\varepsilon, \mathbf{w}, \mathbf{V}, \mathbf{z}) = \mathbf{E}^{-1}(\varepsilon)\mathbf{P}^{-1}\mathbf{c}^{\mathrm{T}}\varphi(\mathbf{w}, \mathbf{V}, \mathbf{z}) \tag{10}$$

such that system nonlinearity $\boldsymbol{f}(\boldsymbol{z})$ fulfills approximation condition

$$\mathbf{f}(\mathbf{z}) = \Phi(\varepsilon, \mathbf{w}, \mathbf{V}, \mathbf{z}), \tag{11}$$

where

$$\mathbf{E}(\varepsilon) = \mathrm{diag}\left(\begin{bmatrix} \varepsilon & \varepsilon^2 & \cdots & \varepsilon^N \end{bmatrix}\right), \tag{12}$$

$$\varphi(\mathbf{w}, \mathbf{V}, \mathbf{z}) = \mathbf{w}^{\mathrm{T}}\mathbf{S}\left(\mathbf{V}^{\mathrm{T}}\mathbf{z}\right), \tag{13}$$

$\varepsilon>0$ is a prescribed positive constant, $\boldsymbol{w}\in\mathrm{R}^l$ and $\boldsymbol{V}\in\mathrm{R}^{n\times l}$ are MLP weight matrices, $\boldsymbol{P}\in\mathrm{R}^{n\times n}$ is a positive-definite symmetric matrix to be determined, $\boldsymbol{c}$ denotes the system output vector defined by (9), and $\boldsymbol{S}$: $\mathrm{R}^n\to\mathrm{R}^n$ is the vector-valued activation function satisfying

$$\mathbf{S}\left(\mathbf{V}^{\mathrm{T}}\mathbf{z}\right) = \begin{bmatrix} s\left(\mathbf{v}_1^{\mathrm{T}}\mathbf{z}\right) & s\left(\mathbf{v}_2^{\mathrm{T}}\mathbf{z}\right) & \cdots & s\left(\mathbf{v}_l^{\mathrm{T}}\mathbf{z}\right) \end{bmatrix}^{\mathrm{T}} \tag{14}$$

with

$$s(x) = \frac{1}{1+e^{-x}}, \quad \forall x \in \mathrm{R}, \tag{15}$$

being the sigmoid function and vector $\boldsymbol{v}_i$ being the ith column of weight matrix $\boldsymbol{V}$ ($i=1, \ldots, l$). Since nonlinear function $\boldsymbol{f}(\boldsymbol{z})$ is norm-bounded, weights $\boldsymbol{w}$ and $\boldsymbol{V}$ are also norm-bounded.

For the above extended system, design the high order neural network extended state observer (HoNN-ESO) as

$$\dot{\widehat{\mathbf{z}}} = \mathbf{A}_{\mathrm{o}}\widehat{\mathbf{z}} + \mathbf{E}^{-1}(\varepsilon)\left[\mathbf{P}^{-1}\mathbf{c}^{\mathrm{T}}\varphi(\widehat{\mathbf{w}}, \widehat{\mathbf{V}}, \widehat{\mathbf{z}}) - \mathbf{l}(\widehat{y} - y)\right], \tag{16}$$

where

$$\varphi(\widehat{\mathbf{w}}, \widehat{\mathbf{V}}, \widehat{\mathbf{z}}) = \widehat{\mathbf{w}}^{\mathrm{T}}\mathbf{S}\left(\widehat{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right), \tag{17}$$

$$\widehat{y} = \mathbf{c}\widehat{\mathbf{z}}, \tag{18}$$

$$\mathbf{l} = [\alpha_1 \quad \alpha_2 \quad \cdots \quad \alpha_N]^{\mathrm{T}}, \tag{19}$$

$\widehat{\mathbf{z}} \in \mathrm{R}^N$ is the observation of extended state-vector $\boldsymbol{z}, \widehat{\mathbf{w}} \in \mathrm{R}^l$ and $\widehat{\mathbf{V}} \in \mathrm{R}^{N\times l}$ are weight estimations of weightings $\boldsymbol{w}$ and $\boldsymbol{V}$ respectively, and α_i ($i = 1, \ldots, N$) are selected such that algebraic equation

$$s^N + \alpha_1 s^{N-1} + \cdots + \alpha_{N-1}s + \alpha_N = 0 \tag{20}$$

is Hurwitz, namely all the roots of (20) have strictly negative real parts. Since estimated weighs $\widehat{\boldsymbol{w}}$ and $\widehat{\boldsymbol{V}}$ dominate the evolution of extended state estimation, a dedicated learning law must be designed to guarantee that $\widehat{\mathbf{z}}$converges to $\boldsymbol{z}$ asymptotically within a prescribed bounded domain under given conditions.

3. Learning law with convergence analysis

This section derives the online weight-update learning law and proves globally asymptotically bounded estimation over a defined set. To quantitatively characterize observer convergence, the observation error is defined as:

$$\widetilde{\mathbf{z}} = \widehat{\mathbf{z}} - \mathbf{z}. \tag{21}$$

Subtracting (16) from (1), the observation error dynamics can be given by

$$\dot{\widetilde{\mathbf{z}}} = \left[\mathbf{A}_{\mathrm{o}} - \mathbf{E}^{-1}(\varepsilon)\mathbf{l}\mathbf{c}\right]\widetilde{\mathbf{z}} + \mathbf{E}^{-1}(\varepsilon)\mathbf{P}^{-1}\mathbf{c}^{\mathrm{T}}[\varphi(\widehat{\mathbf{W}}, \widehat{\mathbf{V}}, \widehat{\mathbf{z}}) - \varphi(\mathbf{W}, \mathbf{V}, \mathbf{z})] - \mathbf{d}. \tag{22}$$

In addition, by considering coordinate transformation

$$\widetilde{\omega} = \frac{1}{\varepsilon}\mathbf{E}(\varepsilon)\widetilde{\mathbf{z}} \tag{23}$$

and scale transformation

$$\tau = \frac{t}{\varepsilon} \tag{24}$$

where t is time, it can be derived that observation error dynamics (22) can be rewritten as

$$\frac{\mathrm{d}\widetilde{\omega}}{\mathrm{d}\tau} = \mathbf{A}\widetilde{\omega} + \mathbf{P}^{-1}\mathbf{c}^{\mathrm{T}}[\varphi(\widehat{\mathbf{w}}, \widehat{\mathbf{V}}, \widehat{\mathbf{z}}) - \varphi(\mathbf{w}, \mathbf{V}, \mathbf{z})] - \varepsilon^N\mathbf{d}, \tag{25}$$

where

$$\mathbf{A} = \mathbf{A}_{\mathrm{o}} - \mathbf{l}\mathbf{c} = \begin{bmatrix} -\alpha_1 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ -\alpha_{N-1} & 0 & \cdots & 1 \\ -\alpha_N & 0 & \cdots & 0 \end{bmatrix}. \tag{26}$$

In the following part of this section, a learning law of weight estimates $\widehat{\boldsymbol{w}}$ and $\widehat{\boldsymbol{V}}$ is given for online updating, so as to provide globally asymptotically bounded convergence of observer (16) on a defined set under specified condition. Before proposing the main result about online learning, a useful lemma is first introduced.

Lemma. The linearly parameterized approximation error of function (17) with respect to function (13) is given by

$$\varphi(\widehat{\mathbf{w}}, \widehat{\mathbf{V}}, \widehat{\mathbf{z}}) - \varphi(\mathbf{w}, \mathbf{V}, \mathbf{z}) = \widetilde{\mathbf{w}}^{\mathrm{T}}\left(\widehat{\mathbf{S}} - \widehat{\mathbf{S}}'\widehat{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right) + \widehat{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}} + d_{\mathrm{a}}, \tag{27}$$

with approximation error d_{a} satisfying

$$d_{\mathrm{a}} \le \frac{l}{4}(\| \widetilde{\mathbf{w}} \|_2 \| \mathbf{V} \|_{\mathrm{F}} + \| \mathbf{w} \|_2 \| \widetilde{\mathbf{V}} \|_{\mathrm{F}}) \| \widehat{\mathbf{z}} \|_2 + 2D_{\mathrm{Z}}(\mathbf{W}), \tag{28}$$

where

$$\widetilde{\mathbf{w}} = \widehat{\mathbf{w}} - \mathbf{w}, \tag{29}$$

$$\widetilde{\mathbf{V}} = \widehat{\mathbf{V}} - \mathbf{V}, \tag{30}$$

$$\mathbf{S}'(\chi) = \begin{bmatrix} s'(\chi_1) & & \\ & \ddots & \\ & & s'(\chi_l) \end{bmatrix}, \quad \forall \chi = \begin{bmatrix} \chi_1 \\ \vdots \\ \chi_l \end{bmatrix} \in \mathrm{R}^l, \tag{31}$$

$$s'(v) = \frac{\mathrm{d}s(v)}{\mathrm{d}v} = \frac{\mathrm{d}}{\mathrm{d}v}\left(\frac{1}{1+e^{-v}}\right) = \frac{e^{-v}}{(1+e^{-v})^2}, \quad \forall v \in \mathrm{R}, \tag{32}$$

$$D_{\mathrm{z}}(\mathbf{w}) = \sum_{j=1}^{l} |w_j|, \tag{33}$$

$||\bullet||_{\mathrm{F}}$ is the Frobenius norm defined as

$$\| \mathbf{R} \|_{\mathrm{F}}^2 = \mathrm{tr}\left(\mathbf{R}^{\mathrm{T}}\mathbf{R}\right), \quad \forall \mathbf{R} \in \mathrm{R}^{n_1 \times n_2}, \quad n_1, n_2 \in \mathrm{N}. \tag{34}$$

Proof. By adding and subtracting $\varphi(\mathbf{w}, \mathbf{V}, \widehat{\mathbf{z}})$, we have

$$\varphi(\widehat{\mathbf{w}}, \widehat{\mathbf{V}}, \widehat{\mathbf{z}}) - \varphi(\mathbf{w}, \mathbf{V}, \mathbf{z}) = \varphi(\widehat{\mathbf{w}}, \widehat{\mathbf{V}}, \widehat{\mathbf{z}}) - \varphi(\mathbf{w}, \mathbf{V}, \widehat{\mathbf{z}}) + d_{\mathrm{z}}, \tag{35}$$

where

$$d_{\mathrm{z}} = \varphi(\mathbf{w}, \mathbf{V}, \widehat{\mathbf{z}}) - \varphi(\mathbf{w}, \mathbf{V}, \mathbf{z}) = \mathbf{w}^{\mathrm{T}}\left[\mathbf{S}\left(\mathbf{V}^{\mathrm{T}}\widehat{\mathbf{z}}\right) - \mathbf{S}\left(\mathbf{V}^{\mathrm{T}}\mathbf{z}\right)\right]. \tag{36}$$

From the definition of sigmoid function (15), all the entries of vector$\mathbf{S}\left(\mathbf{V}^{\mathrm{T}}\widehat{\mathbf{z}}\right) - \mathbf{S}\left(\mathbf{V}^{\mathrm{T}}\mathbf{z}\right)$are bounded by 1, giving that

$$d_{\mathrm{z}} \le D_{\mathrm{z}}(\mathbf{w}). \tag{37}$$

In addition, the Taylor expansion of $\mathbf{S}\left(\mathbf{V}^{\mathrm{T}}\widehat{\mathbf{z}}\right)$ at $\widehat{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}$ gives

$$\mathbf{S}\left(\mathbf{V}^{\mathrm{T}}\widehat{\mathbf{z}}\right) = \widehat{\mathbf{S}} - \widehat{\mathbf{S}}'\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}} + o\left(\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right) \tag{38}$$

where $o\left(\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right)$ is the sum of high order terms in Taylor expansion. It can be further derived from (38) that

$$\begin{aligned} \varphi(\widehat{\mathbf{w}}, \widehat{\mathbf{V}}, \widehat{\mathbf{z}}) - \varphi(\mathbf{w}, \mathbf{V}, \widehat{\mathbf{z}}) &= \widehat{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}} - (\widehat{\mathbf{w}} - \widetilde{\mathbf{w}})^{\mathrm{T}}\left[\widehat{\mathbf{S}} - \widehat{\mathbf{S}}'\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}} + o\left(\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right)\right] \\ &= \widetilde{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}} + \widehat{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}} - \widetilde{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'(\widehat{\mathbf{V}} - \mathbf{V})^{\mathrm{T}}\widehat{\mathbf{z}} - \mathbf{w}^{\mathrm{T}}o\left(\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right) \\ &= \widetilde{\mathbf{w}}^{\mathrm{T}}\left(\widehat{\mathbf{S}} - \widehat{\mathbf{S}}'\widehat{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right) + \widehat{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}} + d_{\mathrm{u}}, \end{aligned} \tag{39}$$

where

$$d_{\mathrm{u}} = \widetilde{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'\mathbf{V}^{\mathrm{T}}\widehat{\mathbf{z}} - \mathbf{w}^{\mathrm{T}}o\left(\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right). \tag{40}$$

As seen from (40) and (38),

$$\begin{aligned} d_{\mathrm{u}} &= \widetilde{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'\mathbf{V}^{\mathrm{T}}\widehat{\mathbf{z}} - \mathbf{w}^{\mathrm{T}}\left[\mathbf{S}\left(\mathbf{V}^{\mathrm{T}}\widehat{\mathbf{z}}\right) - \widehat{\mathbf{S}} + \widehat{\mathbf{S}}'\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right] \\ &= \widetilde{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'\mathbf{V}^{\mathrm{T}}\widehat{\mathbf{z}} - \mathbf{w}^{\mathrm{T}}\widehat{\mathbf{S}}'\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}} + \mathbf{w}^{\mathrm{T}}\left[\widehat{\mathbf{S}} - \mathbf{S}\left(\mathbf{V}^{\mathrm{T}}\widehat{\mathbf{z}}\right)\right], \end{aligned} \tag{41}$$

giving that

$$\begin{aligned} d_{\mathrm{u}} &\le \left|\widetilde{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'\mathbf{V}^{\mathrm{T}}\widehat{\mathbf{z}}\right| + \left|\mathbf{w}^{\mathrm{T}}\widehat{\mathbf{S}}'\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right| + \left|\mathbf{w}^{\mathrm{T}}\left[\widehat{\mathbf{S}} - \mathbf{S}\left(\mathbf{V}^{\mathrm{T}}\widehat{\mathbf{z}}\right)\right]\right| \\ &\le (\| \widetilde{\mathbf{w}} \|_2 \| \mathbf{V} \|_{\mathrm{F}} + \| \mathbf{w} \|_2 \| \widetilde{\mathbf{V}} \|_{\mathrm{F}}) \| \widehat{\mathbf{S}}' \|_{\mathrm{F}} \| \widehat{\mathbf{z}} \|_2 + D_{\mathrm{Z}}(\mathbf{w}). \end{aligned} \tag{42}$$

It is easy to check from (32) that $s'(v)$ reaches its maximal value 0.25 at $v = 0$, which determines that

$$\| \widehat{\mathbf{S}}' \|_{\mathrm{F}}^2 = \mathrm{tr}\left\{(\widehat{\mathbf{S}}')^{\mathrm{T}}\widehat{\mathbf{S}}'\right\} \le \frac{l^2}{16}. \tag{43}$$

Then, we can further draw from inequalities (42) and (43) that

$$d_{\mathrm{u}} \leq \frac{l}{4}(\| \widetilde{\mathbf{w}} \|_2 \| \mathbf{V} \|_{\mathrm{F}} + \| \mathbf{w} \|_2 \| \widetilde{\mathbf{V}} \|_{\mathrm{F}}) \| \widehat{\mathbf{z}} \|_2 + D_{\mathrm{Z}}(\mathbf{w}). \tag{44}$$

Based on (39) and (35), vector d_{a} is given by

$$d_{\mathrm{a}} = d_{\mathrm{u}} + d_{\mathrm{z}}, \tag{45}$$

and from inequalities (37) and (42),

$$d_{\mathrm{a}} = d_{\mathrm{u}} + d_{\mathrm{z}} \leq \frac{l}{4}(\| \widetilde{\mathbf{w}} \|_2 \| \mathbf{V} \|_{\mathrm{F}} + \| \mathbf{w} \|_2 \| \widetilde{\mathbf{V}} \|_{\mathrm{F}}) \| \widehat{\mathbf{z}} \|_2 + 2D_{\mathrm{Z}}(\mathbf{W}), \tag{46}$$

which completes the proof of this lemma.

With the result from the lemma, observation error dynamics (22) can be rewritten as

$$\frac{\mathrm{d}\widetilde{\omega}}{\mathrm{d}\tau} = \mathbf{A}\widetilde{\omega} + \mathbf{P}^{-1}\mathbf{c}^{\mathrm{T}}\left[\widetilde{\mathbf{w}}^{\mathrm{T}}\left(\widehat{\mathbf{S}} - \widehat{\mathbf{S}}'\widehat{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right) + \widehat{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}} + d_{\mathrm{a}}\right]. \tag{47}$$

Accordingly. The main theorem below provides the learning law of weight estimates $\widehat{\boldsymbol{w}}$ and $\widehat{\boldsymbol{V}}$ to ensure asymptotically bounded stability of error system (47).

Theorem. Suppose gain vector $\boldsymbol{l}$ defined by (19) is designed so that system matrix $\boldsymbol{A}$ satisfying (26) is Hurwitz. HoNN-ESO (16) is globally asymptotically convergent with a bounded observation error on the set

$$\mathrm{S}_{\mathrm{w,V}} = \{\widetilde{\mathbf{w}}, \widetilde{\mathbf{V}} | \| \widetilde{\mathbf{w}} \|_2 \leq \mu \| \mathbf{w} \|_2, \| \widetilde{\mathbf{V}} \|_{\mathrm{F}} \leq \mu \| \mathbf{V} \|_{\mathrm{F}}\}, \tag{48}$$

if weighting estimations $\widehat{\boldsymbol{w}}$ and $\widehat{\boldsymbol{V}}$ are adapted by learning law

$$\begin{cases} \dot{\widehat{\mathbf{w}}} = -\varepsilon^{-1}\boldsymbol{\Gamma}_{\mathrm{w}}\left[\widetilde{y}\left(\widehat{\mathbf{S}} - \widehat{\mathbf{S}}'\widehat{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right) + \sigma_{\mathrm{w}}\widehat{\mathbf{w}}\right], \\ \dot{\widehat{\mathbf{V}}} = -\varepsilon^{-1}\boldsymbol{\Gamma}_{\mathrm{V}}\left(\widetilde{y}\widehat{\mathbf{z}}\widehat{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}' + \sigma_{\mathrm{V}}\widehat{\mathbf{V}}\right), \end{cases} \tag{49}$$

where $\boldsymbol{\Gamma}_{\mathrm{w}}$ and $\boldsymbol{\Gamma}_{\mathrm{V}}$ are given constant positive-definite symmetric matrices, and μ, σ_{w} and σ_{V} are all given positive constants, and

$$\widetilde{y} = \widehat{y} - y \tag{50}$$

is the estimation error of system output.

Proof. As system matrix $\boldsymbol{A}$ defined by (26) is Hurwitz, positive-definite symmetric matrix $\boldsymbol{P}$ in (16) solves Lyapunov equation

$$\mathbf{A}^{\mathrm{T}}\mathbf{P} + \mathbf{P}\mathbf{A} = -\mathbf{Q}, \tag{51}$$

where $\boldsymbol{Q}$ is an arbitrarily given positive-definite symmetric matrix, and system matrix $\boldsymbol{A}$ is defined by (26). Choose the Lyapunov function candidate of NN observer (16) as

$$V(\widetilde{\omega}, \widetilde{\mathbf{w}}, \widetilde{\mathbf{V}}) = \frac{1}{2}\left[\widetilde{\omega}^{\mathrm{T}}\mathbf{P}\widetilde{\omega} + \widetilde{\mathbf{w}}^{\mathrm{T}}\boldsymbol{\Gamma}_{\mathrm{w}}^{-1}\widetilde{\mathbf{w}} + \mathrm{tr}\left\{\widetilde{\mathbf{V}}^{\mathrm{T}}\boldsymbol{\Gamma}_{\mathrm{V}}^{-1}\widetilde{\mathbf{V}}\right\}\right]. \tag{52}$$

Differentiate function V defined by (52) along the trajectory of observation error dynamics (47),

$$\begin{aligned} \frac{\mathrm{d}V(\widetilde{\omega}, \widetilde{\mathbf{w}}, \widetilde{\mathbf{V}})}{\mathrm{d}\tau} &= \widetilde{\omega}^{\mathrm{T}}\mathbf{P}\frac{\mathrm{d}\widetilde{\omega}}{\mathrm{d}\tau} + \widetilde{\mathbf{w}}^{\mathrm{T}}\boldsymbol{\Gamma}_{\mathrm{w}}^{-1}\frac{\mathrm{d}\widetilde{\mathbf{w}}}{\mathrm{d}\tau} + \mathrm{tr}\left\{\widetilde{\mathbf{V}}^{\mathrm{T}}\boldsymbol{\Gamma}_{\mathrm{V}}^{-1}\frac{\mathrm{d}\widehat{\mathbf{V}}}{\mathrm{d}\tau}\right\} \\ &= \frac{1}{2}\widetilde{\omega}^{\mathrm{T}}\left(\mathbf{A}^{\mathrm{T}}\mathbf{P} + \mathbf{P}\mathbf{A}\right)\widetilde{\omega} + \widetilde{y}d_{\mathrm{a}} + \varepsilon^{N}\widetilde{\omega}^{\mathrm{T}}\mathbf{P}\mathbf{d} + \widetilde{y}\left[\widetilde{\mathbf{w}}^{\mathrm{T}}\left(\widehat{\mathbf{S}} - \widehat{\mathbf{S}}'\widehat{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right) + \widehat{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right] + \widetilde{\mathbf{w}}^{\mathrm{T}}\boldsymbol{\Gamma}_{\mathrm{w}}^{-1}\frac{\mathrm{d}\widetilde{\mathbf{w}}}{\mathrm{d}\tau} + \mathrm{tr}\left\{\widetilde{\mathbf{V}}^{\mathrm{T}}\boldsymbol{\Gamma}_{\mathrm{V}}^{-1}\frac{\mathrm{d}\widehat{\mathbf{V}}}{\mathrm{d}\tau}\right\} \\ &= -\frac{1}{2}\widetilde{\omega}^{\mathrm{T}}\mathbf{Q}\widetilde{\omega} + \widetilde{y}d_{\mathrm{a}} + \varepsilon^{N}\widetilde{\omega}^{\mathrm{T}}\mathbf{P}\mathbf{d} + \mathrm{tr}\left\{\widetilde{\mathbf{w}}^{\mathrm{T}}\boldsymbol{\Gamma}_{\mathrm{w}}^{-1}\left[\varepsilon\dot{\widehat{\mathbf{w}}} + \boldsymbol{\Gamma}_{\mathrm{w}}\left(\widehat{\mathbf{S}} - \widehat{\mathbf{S}}'\widehat{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right)\widetilde{y}\right]\right\} + \mathrm{tr}\left\{\widetilde{\mathbf{V}}^{\mathrm{T}}\boldsymbol{\Gamma}_{\mathrm{V}}^{-1}\left(\varepsilon\dot{\widehat{\mathbf{V}}} + \boldsymbol{\Gamma}_{\mathrm{V}}\widehat{\mathbf{z}}\widehat{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'\widetilde{y}\right)\right\}. \end{aligned} \tag{53}$$

Submitting learning law (49) to (53), one can found that

$$\begin{aligned} \frac{\mathrm{d}V(\widetilde{\omega}, \widetilde{\mathbf{w}}, \widetilde{\mathbf{V}})}{\mathrm{d}\tau} &= -\frac{1}{2}\widetilde{\omega}^{\mathrm{T}}\mathbf{Q}\widetilde{\omega} + \varepsilon^{N}\widetilde{\omega}^{\mathrm{T}}\mathbf{P}\mathbf{d} + \widetilde{y}d_{\mathrm{a}} - \sigma_{\mathrm{w}}\widetilde{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{w}} - \sigma_{\mathrm{V}}\mathrm{tr}\left\{\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{V}}\right\} \\ &= -\frac{1}{2}\widetilde{\omega}^{\mathrm{T}}\mathbf{Q}\widetilde{\omega} - \frac{\sigma_{\mathrm{w}}}{2}\left(\| \widetilde{\mathbf{w}} \|_2^2 + \| \widehat{\mathbf{w}} \|_2^2 - \| \mathbf{w} \|_2^2\right) - \frac{\sigma_{\mathrm{V}}}{2}\left(\| \widetilde{\mathbf{V}} \|_{\mathrm{F}}^2 + \| \widehat{\mathbf{V}} \|_{\mathrm{F}}^2 - \| \mathbf{V} \|_{\mathrm{F}}^2\right) + \widetilde{y}d_{\mathrm{a}} + \varepsilon^{N}\widetilde{\omega}^{\mathrm{T}}\mathbf{P}\mathbf{d} \\ &\leq -\frac{1}{4}\widetilde{\omega}^{\mathrm{T}}\left[\mathbf{Q} - \mathbf{c}\mathbf{c}^{\mathrm{T}}\right]\widetilde{\omega} - \frac{\sigma_{\mathrm{w}}}{2}\| \widetilde{\mathbf{w}} \|_2^2 - \frac{\sigma_{\mathrm{V}}}{2}\| \widetilde{\mathbf{V}} \|_{\mathrm{F}}^2 + \frac{\sigma_{\mathrm{w}}}{2}\| \mathbf{w} \|_2^2 + \frac{\sigma_{\mathrm{V}}}{2}\| \mathbf{V} \|_{\mathrm{F}}^2 + d_{\mathrm{a}}^2 + \varepsilon^{2N}\| \mathbf{d} \|_{\mathbf{P}\mathbf{Q}^{-1}\mathbf{P}}^2, \end{aligned} \tag{54}$$

where

$$\| \mathbf{d} \|_{\mathbf{P}\mathbf{Q}^{-1}\mathbf{P}}^2 = \mathbf{d}^{\mathrm{T}}\mathbf{P}\mathbf{Q}^{-1}\mathbf{P}\mathbf{d}. \tag{55}$$

From inequality (28), it can be seen that

$$d_{\mathrm{a}}^2 \leq \frac{3l^2}{16}\left(\| \widetilde{\mathbf{w}} \|_2^2 \| \mathbf{V} \|_{\mathrm{F}}^2 + \| \mathbf{w} \|_2^2 \| \widetilde{\mathbf{V}} \|_{\mathrm{F}}^2\right)\left(\| \mathbf{z} \|_2^2 + \| \widetilde{\mathbf{z}} \|_2^2\right) + 12D_{\mathrm{Z}}^2(\mathbf{W}). \tag{56}$$

Substitute (56) to (54),

$$\begin{aligned} \frac{\mathrm{d}V(\widetilde{\omega}, \widetilde{\mathbf{w}}, \widetilde{\mathbf{V}})}{\mathrm{d}\tau} \leq -\frac{1}{4}\widetilde{\omega}^{\mathrm{T}}&\left[\mathbf{Q} - \mathbf{c}\mathbf{c}^{\mathrm{T}} - \frac{3l^2\mu^2\varepsilon^2}{4}\left(\| \mathbf{w} \|_2^2 + \| \mathbf{V} \|_{\mathrm{F}}^2\right)\mathbf{E}^{-2}(\varepsilon)\right]\widetilde{\omega} - \frac{1}{2}\left(\sigma_{\mathrm{w}} - \frac{3l^2}{8}\| \mathbf{V} \|_{\mathrm{F}}^2\| \mathbf{z} \|_2^2\right)\| \widetilde{\mathbf{w}} \|_2^2 + 12D_{\mathrm{Z}}^2(\mathbf{W}) \\ &-\frac{1}{2}\left(\sigma_{\mathrm{V}} - \frac{3l^2}{8}\| \mathbf{w} \|_2^2\| \mathbf{z} \|_2^2\right)\| \widetilde{\mathbf{V}} \|_{\mathrm{F}}^2 + \varepsilon^{2N}\| \mathbf{d} \|_{\mathbf{P}\mathbf{Q}^{-1}\mathbf{P}}^2. \end{aligned} \tag{57}$$

in set $S_{w,V}$ defined by (48). Since $\boldsymbol{W}$, $\boldsymbol{V}$ and $\boldsymbol{z}$ are norm-bounded, matrices $\boldsymbol{Q}$ and constants σ_W and σ_V can be chosen so that

$$\mathbf{Q} > 2\mathbf{c}\mathbf{c}^{\mathrm{T}} + \frac{3}{2}l^2\mu^2\varepsilon^2\left(\| \mathbf{w} \|_2^2 + \| \mathbf{V} \|_{\mathrm{F}}^2\right)\mathbf{E}^{-2}(\varepsilon), \tag{58}$$

$$\sigma_W > \frac{3}{4}l^2\varepsilon^2\| \mathbf{V} \|_{\mathrm{F}}^2\| \mathbf{z} \|_2^2, \tag{59}$$

$$\sigma_V > \frac{3}{4}l^2\varepsilon^2\| \mathbf{w} \|_2^2\| \mathbf{z} \|_2^2. \tag{60}$$

Then, from inequalities (57)-(60), it can be obtained that

$$\begin{aligned}\frac{\mathrm{d}V(\widetilde{\omega},\widetilde{\mathbf{w}},\widetilde{\mathbf{V}})}{\mathrm{d}\tau} \le & -\frac{1}{8}\left(\widetilde{\omega}^{\mathrm{T}}\mathbf{Q}\widetilde{\omega} + 2\delta_W\| \widetilde{\mathbf{w}} \|_{\mathrm{F}}^2 + 2\delta_V\| \widetilde{\mathbf{V}} \|_{\mathrm{F}}^2\right) \\ & + \frac{1}{2}\left(\delta_W\| \mathbf{w} \|_{\mathrm{F}}^2 + \delta_V\| \mathbf{V} \|_{\mathrm{F}}^2\right) + 12D_Z^2(\mathbf{w}) + \varepsilon^{2N}\| \mathbf{d} \|_{\mathbf{P}\mathbf{Q}^{-1}\mathbf{P}}^2,\end{aligned} \tag{61}$$

giving that the observation error of HoNN-ESO (16) converges asymptotically to a bounded error on set $S_{w,V}$. This completes the proof of this theorem.

Remark 1. From (48) and (58), as the minimum eigenvalue $\lambda_{\min}(\boldsymbol{Q})$ of positive-definite matrix $\boldsymbol{Q}$ increases, a larger μ can be chosen so that set $S_{w,V}$ sufficiently covers practical operating scenarios. According to the proof, the globally asymptotically bounded convergence property of HoNN-ESO (16) is preserved regardless of design parameters such as observer gain and neuron quantities. Although theoretical convergence is guaranteed once theorem conditions hold, practical performance indicators including convergence rate are affected by parameter tuning. For single-hidden-layer MLPs, larger numbers of hidden-layer neurons generally deliver stronger approximation capability (Mhaskar, 1993; Pinkus, 1999). As indicated by inequalities (58), (59) and (60), as neuron number l increases, larger σ_W, σ_V and matrix $\boldsymbol{Q}$ with higher $\lambda_{\min}(\boldsymbol{Q})$ are required to maintain convergence guarantees, and matrix $\boldsymbol{P}$ inside HoNN-ESO (16) is solved from Lyapunov Eq. (51). The recommended parameter-tuning workflow is summarized as follows: a) Assign observer gain $\boldsymbol{l}$ defined by (19) to ensure characteristic Eq. (20) is Hurwitz; b) Select neuron number l and positive constant ε, then determine matrix $\boldsymbol{Q}$ along with constants σ_W and σ_V to satisfy inequalities (58), (59) and (60). Two major motivations support embedding MLP (10) within ESO: first, single-hidden-layer MLPs have been validated in pattern recognition and intelligent control applications; second, the quantity of trainable parameters are relatively low, which effectively restricts computational overload.

Remark 2. Define cost function as

$$J(\widetilde{\mathbf{z}}) = \frac{1}{2}\widetilde{\mathbf{z}}^{\mathrm{T}}\mathbf{Q}\widetilde{\mathbf{z}}, \tag{62}$$

and the time-derivative of cost function J can be given by

$$\dot{J}(\widetilde{\mathbf{z}}) = \widetilde{\mathbf{z}}^{\mathrm{T}}\mathbf{P}\dot{\widetilde{\mathbf{z}}} = -\frac{1}{2}\widetilde{\mathbf{z}}^{\mathrm{T}}\mathbf{Q}\widetilde{\mathbf{z}} + \widetilde{y}d_a + \widetilde{y}\left[\widetilde{\mathbf{w}}^{\mathrm{T}}\left(\widehat{\mathbf{S}} - \widehat{\mathbf{S}}'\widehat{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right) + \widehat{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right]. \tag{63}$$

From (63), it can be further obtained gradients

$$\frac{\partial\dot{J}}{\partial\widetilde{\mathbf{w}}} = \left(\widehat{\mathbf{S}} - \widehat{\mathbf{S}}'\widehat{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right)\widetilde{y} \tag{64}$$

and

$$\frac{\partial\dot{J}}{\partial\widetilde{\mathbf{V}}} = \widehat{\mathbf{z}}\widehat{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'\widetilde{y}. \tag{65}$$

Based on (64) and (65), learning law (49) can be rewritten as

$$\begin{cases}\dot{\widehat{\mathbf{w}}} = -\varepsilon^{-1}\boldsymbol{\Gamma}_w\left(\dfrac{\partial\dot{J}}{\partial\widetilde{\mathbf{w}}} + \delta_w\widehat{\mathbf{w}}\right), \\ \dot{\widehat{\mathbf{V}}} = -\varepsilon^{-1}\boldsymbol{\Gamma}_V\left(\dfrac{\partial\dot{J}}{\partial\widetilde{\mathbf{V}}} + \delta_V\widehat{\mathbf{V}}\right),\end{cases} \tag{66}$$

showing that the weighting estimations $\widehat{\boldsymbol{w}}$ and $\widehat{\boldsymbol{V}}$ are essentially adapted by the negative gradients of $\dot{J}$ to accelerate convergence.

Remark 3. The implementation schematic of HoNN-ESO (16) is illustrated in Fig. 2. HoNN-ESO essentially constitutes a recurrent neural network (RNN) given by

$$\begin{cases}\dot{\widehat{\mathbf{z}}} = \mathbf{A}_o\widehat{\mathbf{z}} + \mathbf{E}^{-1}(\varepsilon)\left[\mathbf{P}^{-1}\mathbf{c}^{\mathrm{T}}\varphi(\widehat{\mathbf{w}},\widehat{\mathbf{V}},\widehat{\mathbf{z}}) - \mathbf{l}\widetilde{y}\right], \\ \dot{\widehat{\mathbf{w}}} = -\varepsilon^{-1}\boldsymbol{\Gamma}_w\left[\widetilde{y}\left(\widehat{\mathbf{S}} - \widehat{\mathbf{S}}'\widehat{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right) + \sigma_w\widehat{\mathbf{w}}\right], \\ \dot{\widehat{\mathbf{V}}} = -\varepsilon^{-1}\boldsymbol{\Gamma}_V\left(\widetilde{y}\widehat{\mathbf{z}}\widehat{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}' + \sigma_V\widehat{\mathbf{V}}\right),\end{cases} \tag{67}$$

where$\widetilde{y}$is defined by (50), and function φ is given by Eq. (17). Since MLP (13) approximate system nonlinearity $\boldsymbol{f}(\boldsymbol{z})$ defined by (11), online weight training realizes online identification for system dynamics. Observations for x_2, $\bullet\bullet\bullet$, x_n and $\boldsymbol{\zeta}$ correspond to high-order time-derivatives of output y. Hence, HoNN-ESO outputs can serve as smart indicators for anomaly detection. Practical configuration steps are as follows: (1) Select measurable process variable as system output y=x_1=z_1; (2) Specify system-dynamics order n and disturbance-dynamics order m; (3) Assign observer gain $\boldsymbol{l}$, positive constant ε and the number of hidden-layer neurons for MLP (13).

Remark 4. In real-world engineering, measurement noise or disturbance δ_y commonly corrupts sensor readings, giving that practical

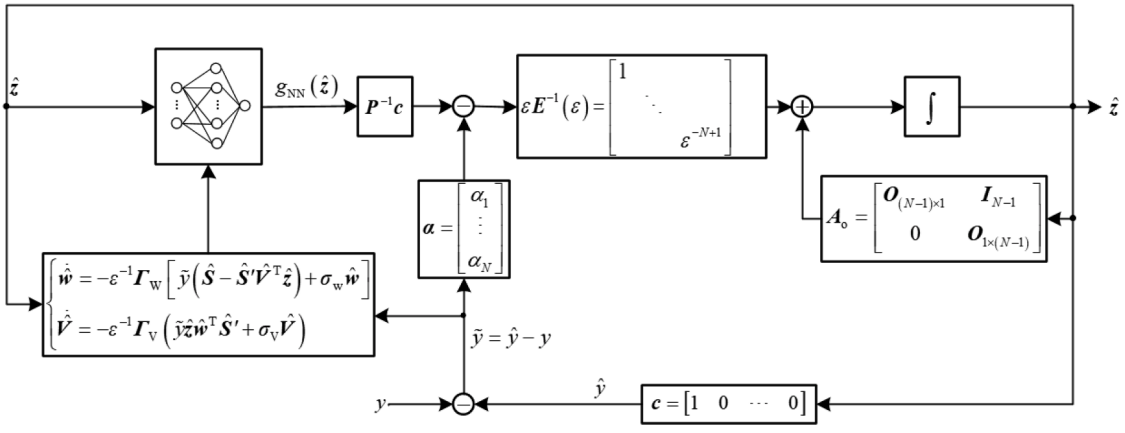


Fig. 2. Schematic diagram for implementing the high order neural network extended state observer (HoNN-ESO).

measured output becomes

$$y_{\mathrm{d}} = y + \delta_y. \tag{68}$$

Replacing y by y_{d} in (16), the HoNN-ESO under disturbance δ_y is given by

$$\dot{\widehat{\mathbf{z}}} = \mathbf{A}_{\mathrm{o}}\widehat{\mathbf{z}} + \mathbf{E}^{-1}(\varepsilon)\left[\mathbf{P}^{-1}\mathbf{c}^{\mathrm{T}}\varphi(\widehat{\mathbf{w}}, \widehat{\mathbf{V}}, \widehat{\mathbf{z}}) - \mathbf{l}(\widehat{y} - y_{\mathrm{d}})\right]. \tag{69}$$

From (21), (23) and (24), we can further obtain that

$$\frac{\mathrm{d}\widetilde{\omega}}{\mathrm{d}\tau} = \mathbf{A}\widetilde{\omega} + \mathbf{P}^{-1}\mathbf{c}^{\mathrm{T}}\left[\widetilde{\mathbf{w}}^{\mathrm{T}}\left(\widehat{\mathbf{S}} - \widehat{\mathbf{S}}'\widehat{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right) + \widehat{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}'\widetilde{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}} + d_{\mathrm{a}}\right] + \mathbf{l}\delta_y, \tag{70}$$

showing disturbance δ_y degrades observation performance. Designing $\boldsymbol{Q}$, σ_{W} and σ_{V} so that inequalities (58), (59) and (60) are satisfied, we can see that

$$\begin{aligned}\frac{\mathrm{d}V(\widetilde{\omega}, \widetilde{\mathbf{w}}, \widetilde{\mathbf{V}})}{\mathrm{d}\tau} &\le -\frac{1}{8}\left(\widetilde{\omega}^{\mathrm{T}}\mathbf{Q}\widetilde{\omega} + 2\sigma_{\mathrm{W}}\| \widetilde{\mathbf{w}} \|_{\mathrm{F}}^2 + 2\sigma_{\mathrm{V}}\| \widetilde{\mathbf{V}} \|_{\mathrm{F}}^2\right) + \frac{1}{2}\left(\delta_{\mathrm{W}}\| \mathbf{W} \|_{\mathrm{F}}^2 + \delta_{\mathrm{V}}\| \mathbf{V} \|_{\mathrm{F}}^2\right) + 12D_{\mathrm{Z}}^2(\mathbf{W}) + \varepsilon^{2N}\| \mathbf{d} \|_{\mathbf{PQ}^{-1}\mathbf{P}}^2 + \widetilde{\omega}^{\mathrm{T}}\mathbf{l}\delta_y \\ &\le -\frac{1}{16}\left(\widetilde{\omega}^{\mathrm{T}}\mathbf{Q}\widetilde{\omega} + 4\sigma_{\mathrm{W}}\| \widetilde{\mathbf{w}} \|_{\mathrm{F}}^2 + 4\sigma_{\mathrm{V}}\| \widetilde{\mathbf{V}} \|_{\mathrm{F}}^2\right) + \frac{1}{2}\left(\delta_{\mathrm{W}}\| \mathbf{W} \|_{\mathrm{F}}^2 + \delta_{\mathrm{V}}\| \mathbf{V} \|_{\mathrm{F}}^2\right) + 12D_{\mathrm{Z}}^2(\mathbf{W}) + \varepsilon^{2N}\| \mathbf{d} \|_{\mathbf{PQ}^{-1}\mathbf{P}}^2 + 4\delta_y^2\mathbf{l}^{\mathrm{T}}\mathbf{Q}^{-1}\mathbf{l}.\end{aligned} \tag{71}$$

From the last term in the right-hand side of (71), it can be seen that the disturbance δ_y deteriorates the convergence of the observer. Conventional ESO relies on high observer gains for convergence and thus is highly sensitive to measurement noise and external disturbances. In contrast, HoNN-ESO mitigates disturbances via online learning from its embedded MLP rather than high-gain amplification.

Remark 5. Network weights $\widehat{\boldsymbol{w}}$ and $\widehat{\boldsymbol{V}}$ act as extra extended state-variables induced by online learning law (49). Given state-vector $\boldsymbol{z}$ and scalar output y, $\widehat{\boldsymbol{w}} \in \mathrm{R}^l$ and $\widehat{\boldsymbol{V}} \in \mathrm{R}^{N\times l}$, producing $(N+1)l$ extended states. Computational burden grows with hidden-layer neuron number l. To reduce computational cost, it is suggested to set $\boldsymbol{\Gamma}_{\mathrm{W}}=\lambda_{\mathrm{W}}\boldsymbol{I}_l$ and $\boldsymbol{\Gamma}_{\mathrm{V}}=\lambda_{\mathrm{V}}\boldsymbol{I}_N$ with positive constants λ_{W} and λ_{V}, which simplifies the observer formulation to be

$$\begin{cases}\dot{\widehat{\mathbf{z}}} = \mathbf{A}_{\mathrm{o}}\widehat{\mathbf{z}} + \mathbf{E}^{-1}(\varepsilon)\left[\mathbf{P}^{-1}\mathbf{c}^{\mathrm{T}}\varphi(\widehat{\mathbf{w}}, \widehat{\mathbf{V}}, \widehat{\mathbf{z}}) - \mathbf{l}\widetilde{y}\right], \\ \dot{\widehat{\mathbf{w}}} = -\varepsilon^{-1}\lambda_{\mathrm{w}}\left[\widetilde{y}\left(\widehat{\mathbf{S}} - \widehat{\mathbf{S}}'\widehat{\mathbf{V}}^{\mathrm{T}}\widehat{\mathbf{z}}\right) + \sigma_{\mathrm{w}}\widehat{\mathbf{w}}\right], \\ \dot{\widehat{\mathbf{V}}} = -\varepsilon^{-1}\lambda_{\mathrm{V}}\left(\widetilde{y}\widehat{\mathbf{z}}\widehat{\mathbf{w}}^{\mathrm{T}}\widehat{\mathbf{S}}' + \sigma_{\mathrm{V}}\widehat{\mathbf{V}}\right).\end{cases} \tag{72}$$

Benefiting from its single-hidden-layer architecture, HoNN-ESO (72) imposes far lower computational complexity compared with deep-neural-network-based alternatives.

Remark 6. NNOs fall into two training paradigms: offline training and online training. For offline-trained NNOs, pure simulation validation cannot provide satisfactory field performance, since simulation-generated training samples inevitably deviate from real-plant dynamics. Additional fine-tuning using real measured operational data is mandatory. To mitigate such tuning overhead, online training schemes with corresponding online learning laws are preferable. As evident from (67) and (72), HoNN-ESO adopts fully online training and does not depend on physical system parameters. Thanks to this model-free merit, bounded mismatches between simulated and real dynamics of industrial process will not destroy its globally asymptotically bounded convergence. Therefore, simulation-based validation is adequate to verify algorithm convergence for HoNN-ESO.

Remark 7. Beyond convergence, online real-time executability is another critical practical concern, predominantly determined by computational complexity. Compared with classical ESO, extra computation of HoNN-ESO arises from its embedded single-hidden-layer MLP, including forward propagation and online weighting adaptation. For HoNN-ESO with state dimension N and l hidden-layer neurons, forward propagation time complexity reads $O(Nl+N^2)$, and the learning-law update contributes $O(Nl)$. The overall computational complexity is $O(Nl+N^2)$, which scales linearly with hidden-layer neuron number l. As demonstrated in Section 4, this complexity level enables real-time online execution.

4. Application to nuclear reactor monitoring

In this section, HoNN-ESO is applied to the monitoring of a modular high temperature gas-cooled reactor (mHTGR), aiming to validate its capacity of generating observations for anomaly detection purposes. Monitoring here refers to the joint observation of system states, lumped disturbances and their time-order derivatives. Numerical simulations under multiple abnormal scenarios are presented with detailed discussions.

4.1. Background of application

The mHTGR is a nuclear fission reactor that employs helium as coolant and graphite as both moderator and structural material. Fuel elements of mHTGR embed thousands of tri-structural isotropic (TRISO)-coated particles within spherical or prismatic graphite matrices. Each TRISO particle features a uranium dioxide kernel encapsulated by multilayer coatings, which can prevent fission-product release up to 1620 °C. Negative temperature feedback effect over the full

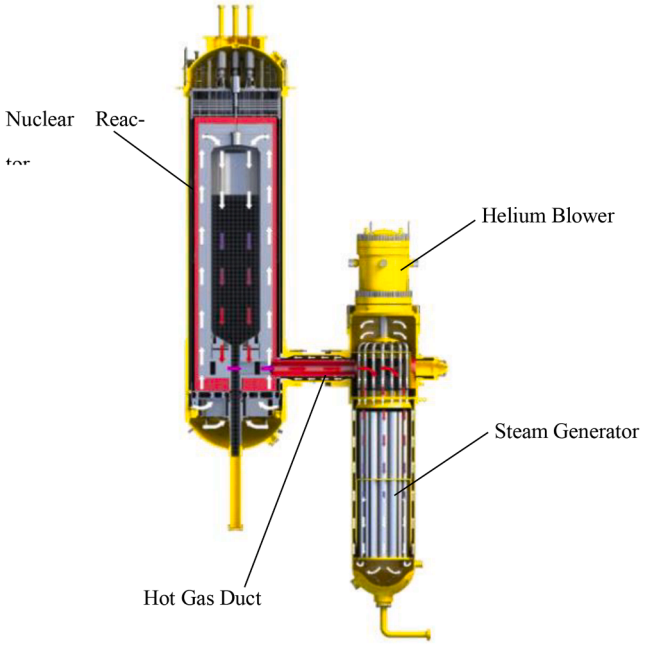


Fig. 3. Schematic diagram of a mHTGR-based reactor module.

power range ensures inherent safety for mHTGR if power density and core diameter are kept below certain thresholds, yielding a single-module rated thermal power around 200MWt. Fig. 3 illustrates the reactor module adopted in HTR-PM and HTR-PM600 plants. Each module consists of an mHTGR, a helical-coil once-through steam generator (OTSG) positioned side-by-side to the reactor, a helium blower mounded above the OTSG, and a hot-gas duct connecting the reactor and steam generator. During operation, helium pressurized by the blower is delivered toward the reactor through the annular pipeline surrounding the hot gas duct. After mixing inside the reactor-pressure-vessel lower plenum, helium flows upward through side-reflector boreholes, reverses direction at the top reflector inner plenum, and travels down through the reactor core. Heated helium is collected and mixed in the bottom-reflector hot-gas plenum, then flows via the hot-gas duct toward the primary side of OTSG, transferring heat to the secondary-side coolant for superheated steam generation. The main steam temperature of mHTGR-based nuclear plants reaches approximately 540 °C (Dong et al., 2018; Zhang et al., 2016; Zhang et al., 2022), considerably higher than typical pressurized water reactor (PWR) plants, whose main-steam temperature stays below 300 °C. Owing to inherent safety advantages (Zhang et al., 2024), mHTGR can not only supply clean electricity to power grids but also deliver large-scale clean heat for district heating and industrial thermochemical processes. For instance, the combined superheated steam outputs from multiple mHTGR nuclear steam supply system (NSSS) modules can support hydrogen production via Cu-Cl thermochemical cycle (Dong, 2022) or methane steam reforming (Li & Dong, 2025).

Limited by single-module thermal-power capacity, multi-modular configurations are widely adopted: superheated steam from several reactor modules is aggregated to drive a single turbine to achieve target power rating. HTR-PM (Dong et al., 2025; Zhang et al., 2016) and HTR-PM600 (Dong et al., 2018; Zhang et al., 2022) are representative multi-modular mHTGR plants, employing two and six reactor modules respectively to drive a shared steam turbine-generator set. Multi-modular layouts require operators to supervise multiple reactor modules simultaneously, which motivates the adoption of state-observation techniques for enhanced operational monitoring. In the remainder of this section, numerical simulations under diverse scenarios are carried out to verify the feasibility of HoNN-ESO for reactor monitoring.

4.2. Simulation results

Numerical simulations are implemented on a lumped-parameter simulation platform for HTR-PM600 plant built within MATLAB/Simulink environment (Zhang et al., 2016). This simulator reproduces major dynamic behavior of the reactor modules, turbine-generator set and balance-of-plant components under power-generation mode. Reactor module dynamics of HTR-PM600 are identical to those of HTR-PM, and the simulator has been calibrated against operation data from the commercially-operational HTR-PM since December 20, 2023 (Dong, 2022). Lumped-parameter model parameters can be conveniently tuned to align simulated responses with real-plant behavior, hence simulation outcomes are reliable for validating HoNN-ESO performance for reactor monitoring.

In the simulation, measurable key process variables including neutron flux, primary flowrate and helium pressure are selected as system output y. System-dynamics order n and disturbance-dynamics m are set as $n{=}m=2$, giving extended-system dimension $N=n{+}m=4$. The positive constant ε is set to be $\varepsilon{=}1.0$, and α_k ($k=1,\ldots,N$) is selected as

$$\alpha_k=\frac{N!a^k}{(N-k)!k!},\quad k=1,\cdots,N, \tag{73}$$

where a is a positive constant determining the poles of system matrix $\boldsymbol{A}$ given by

$$\mathbf{A}=\mathbf{A}_o-\mathbf{lc}, \tag{74}$$

and observer gain $\boldsymbol{l}$ is defined by (19) and (73) accordingly. Different values for positive constant a and MLP hidden-layer-neuron count l are tested to investigate how the embedded MLP influences observation performance. Notably, HoNN-ESO degenerates to canonical ESO

$$\dot{\hat{\mathbf{z}}}=\mathbf{A}_o\hat{\mathbf{z}}-\mathbf{E}^{-1}(\varepsilon)\mathbf{l}\tilde{y}, \tag{75}$$

when hidden-layer-neuron-number l is set to be $l=0$.

Five typical test scenarios are considered in the simulation, including reactivity-step disturbance, helium-flowrate-step disturbance, helium-leakage-rate-step disturbance, measurement-noise disturbance, and measurement-bias disturbance. The simulation results are given as follows.

(1) Case A: Reactivity Stepping Disturbance

The reactivity disturbance is triggered by abnormal control rod insertion, which may deteriorate stable mHTGR operation. The nuclear power plant operates at steady state initially, and a stepping decrease in reactivity of 0.1$ is injected into one reactor module during simulation. Normalized neutron flux is chosen as system output y. Constant a is set to be $a=0.68$, and the poles of system matrix $\boldsymbol{A}$ is given by (73). The response of estimation error $\tilde{y}$ along with the responses of estimations $\hat{x}_2$, $\hat{\zeta}_1$ and $\hat{\zeta}_2$ provided

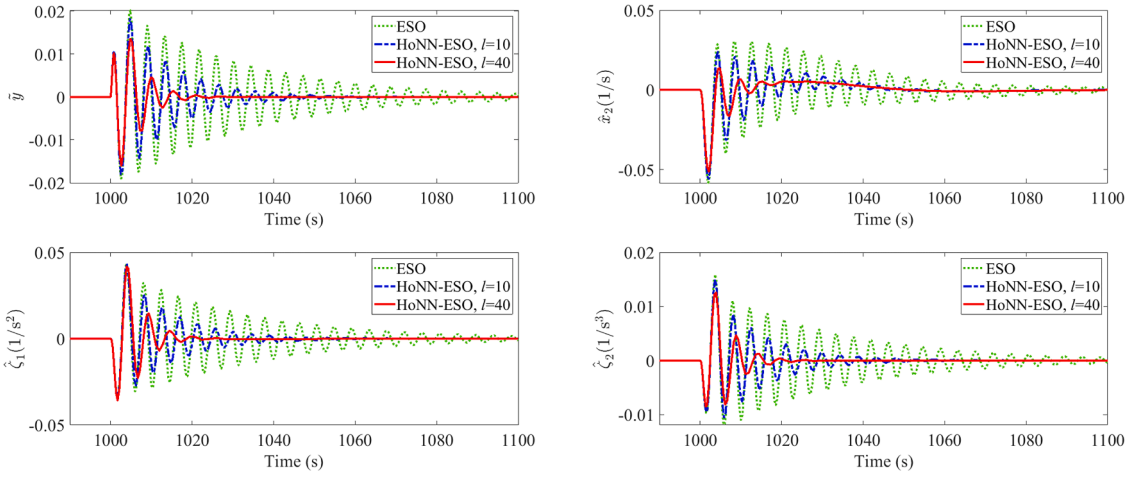


Fig. 4. Responses of estimation error$\tilde{y}$and estimations $\hat{x}_2$, $\hat{\zeta}_1$ and $\hat{\zeta}_2$ with $a=0.68$ in Case A.

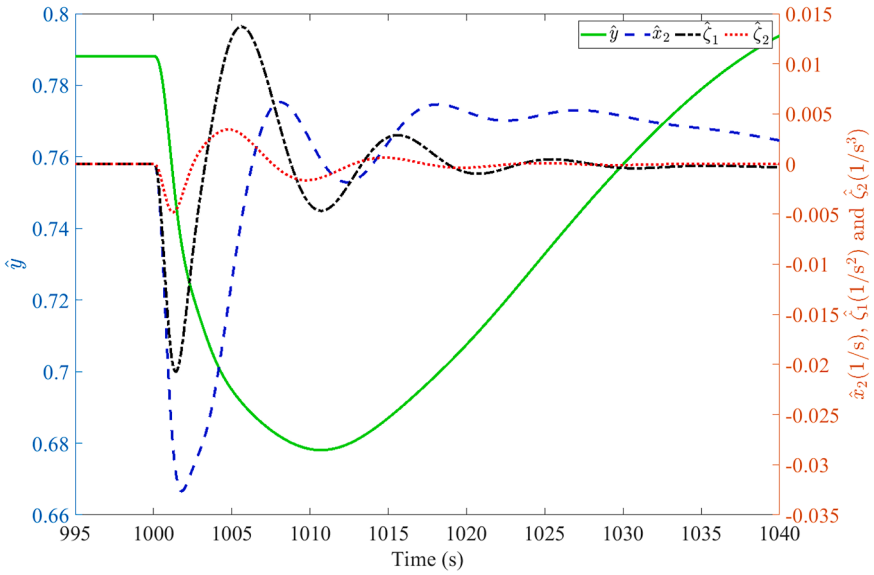


Fig. 5. Responses of $\hat{y}$, $\hat{x}_2$, $\hat{\zeta}_1$ and $\hat{\zeta}_2$ with $a = 0.8$ and $l = 40$ in case A.

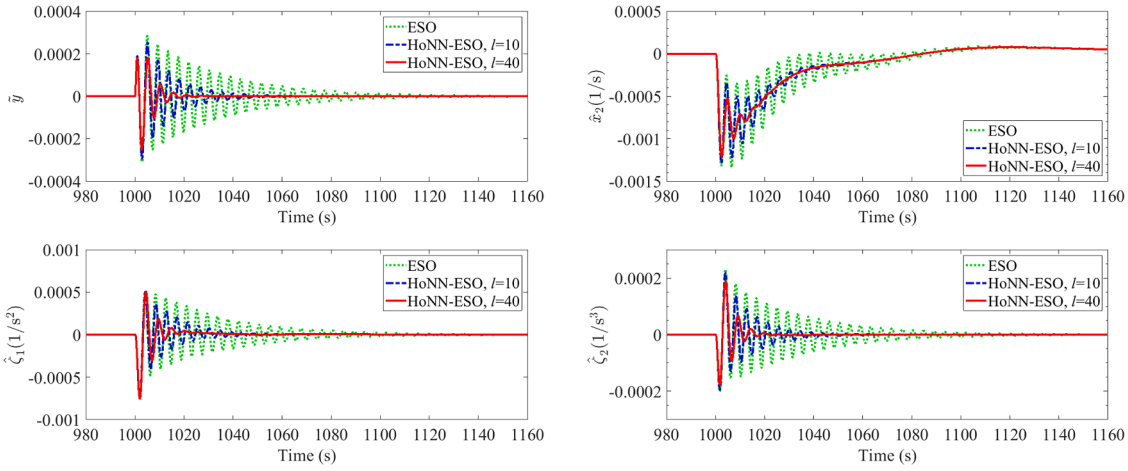


Fig. 6. Responses of estimation error $\tilde{y}$ and estimations $\hat{x}_2$, $\hat{\zeta}_1$ and $\hat{\zeta}_2$ with $a = 0.68$ in Case B.

by both ESO (75) and HoNN-ESO with different hidden-layer neuron number l are shown in Fig. 4. For the setting of $a = 0.8$ and $l = 40$, the responses of reconstructed system output $\hat{y} = \hat{x}_1$ together with observations $\hat{x}_2$, $\hat{\zeta}_1$ and $\hat{\zeta}_2$ obtained from the HoNN-ESO are shown in Fig. 5.

(2) Case B: Helium Flowrate Stepping Disturbance

The helium flowrate disturbance originates from abnormal rotational deceleration of the primary helium blower. The nuclear power plant maintains steady-power operation at the beginning, and a step decrease of 5kg/s in primary helium flowrate is imposed on one reactor module. Normalized neutron flux is still selected as system output y. For constant a being given as $a = 0.68$, the responses of $\tilde{y}$ along with $\hat{x}_2$, $\hat{\zeta}_1$ and $\hat{\zeta}_2$ obtained from both classical ESO (75) and HoNN-ESO with different hidden-layer neuron number l are shown in Fig. 6. For the setting of $a = 0.8$ and $l = 40$, the responses of $\hat{y}$, $\hat{x}_2$, $\hat{\zeta}_1$ and $\hat{\zeta}_2$ provided by HoNN-ESO are given in Fig. 7.

(3) Case C: Helium Leakage-Rate Stepping Disturbance

The plant operates at the given power-level initially, and a step increment of 0.2 kg/s in helium leakage-rate is introduced to one reactor module. Primary helium pressure is selected as system output y. For constant a being set as $a = 0.68$, the responses of output estimation error $\tilde{y}$ along with observations $\hat{x}_2$, $\hat{\zeta}_1$ and $\hat{\zeta}_2$ corresponding to both classical ESO (75) and HoNN-ESO with different hidden-layer neuron numbers are shown in Fig. 8. For $a = 0.8$ and $l = 40$, the responses of observations $\hat{y}$, $\hat{x}_2$, $\hat{\zeta}_1$ and $\hat{\zeta}_2$ provided by HoNN-ESO are shown in Fig. 9.

(4) Case D: Robustness to Measurement Noise

Zero-mean Gaussian white noise with covariance σ^2=1e-6 is superimposed on normalized neutron flux measurements. This noise level is relatively high for nuclear instrumentation systems, with a standard deviation σ equivalent to 0.1 % of rated reactor-full-power (RFP). The plant stays steady initially, and a step reduction of 0.1$ in reactivity is injected. Normalized neutron flux is set as system output y. With constant a being selected as $a = 0.68$ and $a = 0.8$, the response of observation error $\tilde{y}$ along with the responses of $\hat{x}_2$, $\hat{\zeta}_1$ and $\hat{\zeta}_2$ obtained from both classical ESO (75) and HoNN-ESO with different hidden-layer neuron numbers are shown in Figs. 10 and 11.

(5) Case E: Robustness to Measurement Bias

Constant measurement bias of normalized neutron flux caused by sensor faults is investigated for testing robustness. In practice,

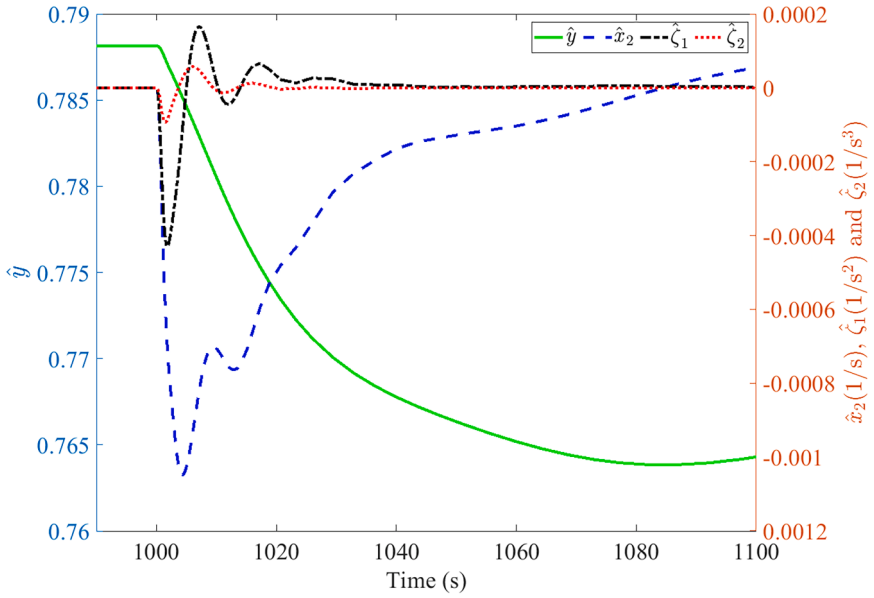


Fig. 7. Responses of $\widehat{y}$, $\widehat{x}_2$, $\widehat{\zeta}_1$ and $\widehat{\zeta}_2$ with $a = 0.8$ and $l = 40$ in case B.

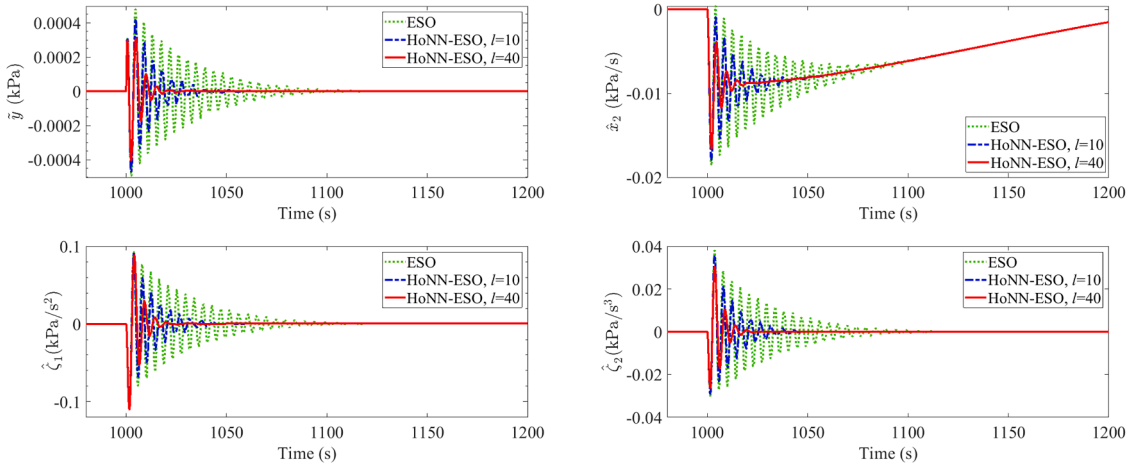


Fig. 8. Responses of estimation error$\tilde{y}$and estimations $\widehat{x}_2$, $\widehat{\zeta}_1$ and $\widehat{\zeta}_2$ with $a = 0.68$ and different l in Case C.

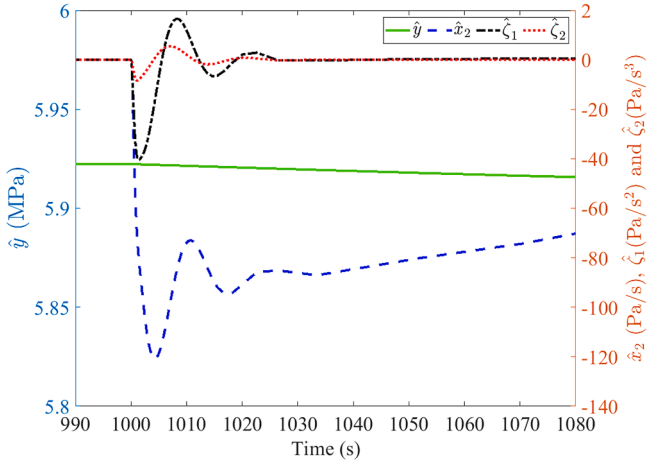


Fig. 9. Responses of $\widehat{y}$, $\widehat{x}_2$, $\widehat{\zeta}_1$ and $\widehat{\zeta}_2$ with $a = 0.8$ and $l = 40$ in case C.

process variable measurements are usually averaged from redundant sensor outputs, and a single-sensor failure may introduce constant bias and degrade observer performance. A constant measurement bias equal to 10 % full-reactor power is added to normalized-neutron-flux readings with $a = 0.8$. Estimation responses for ESO and HoNN-ESO under different neuron counts are given in Fig. 12.

4.3. Discussions

Simulation results from Figs. 4, 6 and 8 verify that HoNN-ESO yields globally asymptotically bounded observations for system output y as well as x_2, ζ_1, and ζ_2 over a defined set under specified conditions. Observation performance is substantially improved by increasing hidden-layer neuron quantity, being consistent with the well-known conclusion that more neutrons enhance neural network learning capability. In our test, constants a and ε are set no greater than 1.0, resulting in relatively low observer gain $\boldsymbol{l}$ given by (19) and (73). Low observer gains bring better immunity against measurement noises but inevitably slow down observation convergence. As shown by the responses of $\widetilde{y}$, $\widehat{x}_2$,

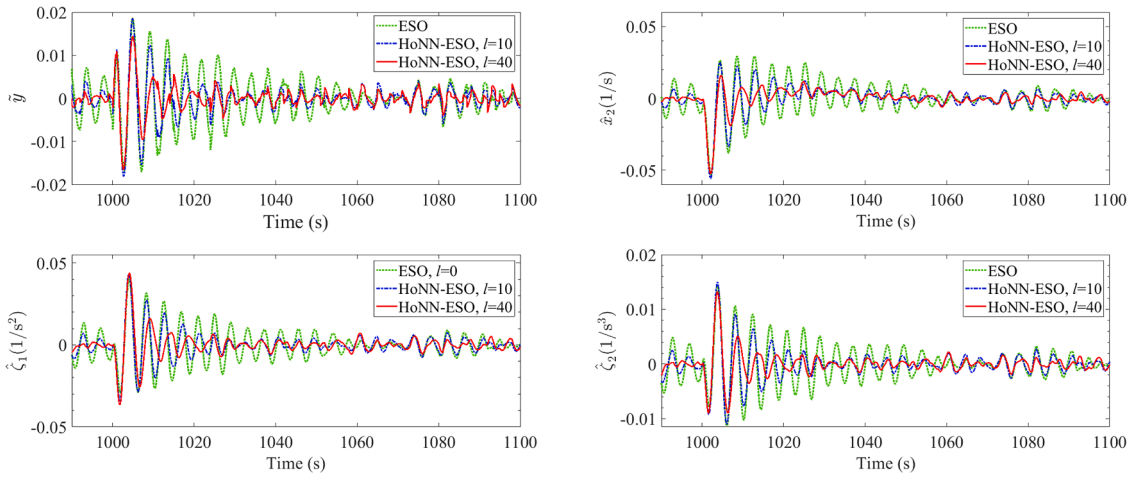


Fig. 10. Responses of estimation error$\tilde{y}$and estimations $\hat{x}_2$, $\hat{\zeta}_1$ and $\hat{\zeta}_2$ with $a=0.68$ and $\sigma^2=1\times10^{-6}$ in Case D.

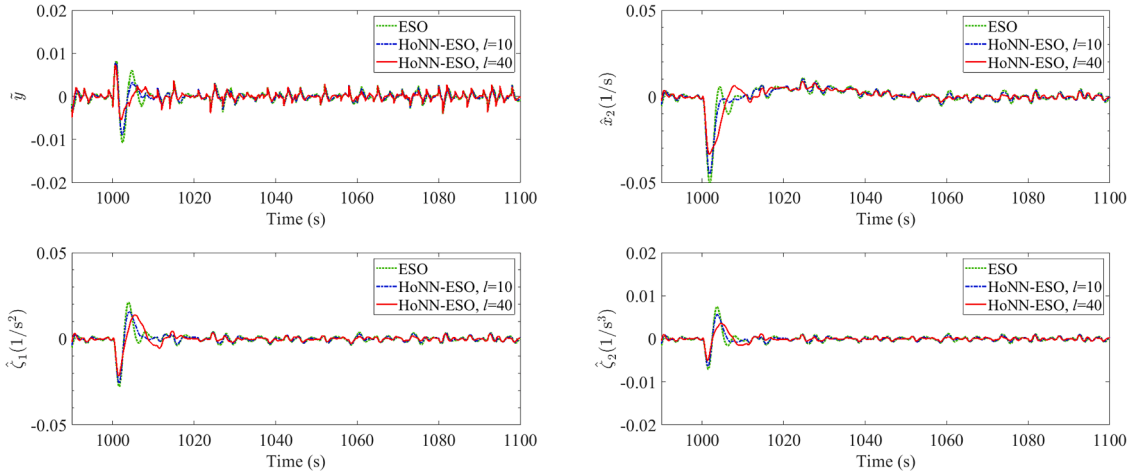


Fig. 11. Responses of estimation error$\tilde{y}$and estimations $\hat{x}_2$, $\hat{\zeta}_1$ and $\hat{\zeta}_2$ with $a=0.8$ and $\sigma^2=1\times10^{-6}$ in Case D.

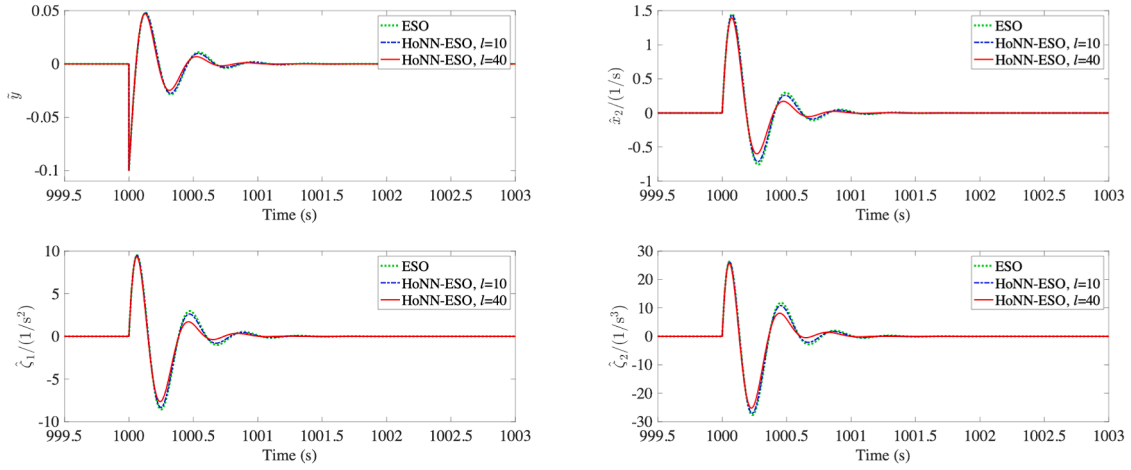


Fig. 12. Responses of estimation error$\tilde{y}$and estimations $\hat{x}_2$, $\hat{\zeta}_1$ and $\hat{\zeta}_2$ with $a=0.8$ and $\sigma^2=1\times10^{-6}$ in Case E.

$\hat{\zeta}_1$ and $\hat{\zeta}_2$ in Figs. 4, 6 and 8, classical ESO exhibits pronounced oscillations, while such oscillations are significantly suppressed by HoNN-ESO. Robustness against measurement bias further improves as neuron count increases. These results confirm the performance gain brought by embedding MLP within the ESO framework.

For the reactivity stepping disturbance induced by abnormal control rod insertion, negative reactivity stepping lower neutron flux and consequently reduce fuel temperature. Doppler-effect-driven negative temperature feedback generates compensating positive reactivity,

Table 2
Pairing of typical anomalies and observed variables based on HoNN-ESO.

Anomalies	System outputs	Observed variables for detection
Reactivity stepping	Normalized neutron flux	$\widehat{\zeta}_2$
Helium flowrate stepping	Normalized neutron flux	$\widehat{\zeta}_1$
Helium leakage	Primary helium pressure	$\widehat{\zeta}_2$

which drives neutron flux to rise until reactivity equilibrium is restored. In Fig. 4, initial growth of observation error $\widetilde{y}$ corresponds to neutron-flux drop, while the subsequent decay reflects temperature reactivity feedback. After short transients, $\widetilde{y}$ converges asymptotically to a bounded neighborhood around zero, indicating that observation $\widehat{y}$ closely match real sensor measurement y. From Fig. 4, observations $\widehat{x}_2$, $\widehat{\zeta}_1$ and $\widehat{\zeta}_2$ in Fig. 4 immediately turn negative once reactivity steps occur, demonstrating their capacity to indicate detecting reactivity-reduction anomalies. As further shown in Fig. 5, $\widehat{\zeta}_2$ representing the time derivative of external disturbance, reaches its minimum value faster than $\widehat{x}_2$ and $\widehat{\zeta}_1$.When normalized neutron flux adopted as system output, $\widehat{\zeta}_2$ is particularly suitable for detecting reactivity-step anomalies.

For the helium-flowrate-step disturbance caused by blower rotational slowdown, step-wise helium-flowrate reduction elevates fuel temperature and introduce negative reactivity that reduces the neutron flux. Decreased neutron flux then cools fuel elements and produces positive reactivity to counterbalance flow-reduction induced negative reactivity. As shown in Fig. 6, initial growth of $\widetilde{y}$ reflects neutron-flux decrease. Observations $\widehat{x}_2$, $\widehat{\zeta}_1$ and $\widehat{\zeta}_2$ shown in Fig. 7 turn negative and capture the flux-dropping event. Moreover, $\widehat{\zeta}_1$ quantifies external-disturbance magnitude and $\widehat{\zeta}_2$ characterizes disturbance-variation trends; both reflect initial degradation and subsequent reactivity recovery. From Fig. 7, $\widehat{\zeta}_2$ responds earlier than $\widehat{x}_2$ and $\widehat{\zeta}_1$. With normalized neutron flux as output, $\widehat{\zeta}_1$ and $\widehat{\zeta}_2$ serve as effective indicators for helium-flow-step anomalies.

For helium-leakage-rate-step scenarios, primary helium pressure decreases gradually after leakage occurs. Since leakage-rates are generally small, pressure fluctuations induced by helium-temperature variation may mask leakage signatures and delay operator detection. From Fig. 8, $\widehat{x}_2$, $\widehat{\zeta}_1$ and $\widehat{\zeta}_2$ decrease instantaneously upon leakage onset and effectively capture such events. As shown in Fig. 9, $\widehat{\zeta}_2$ attains its minimum ahead of both $\widehat{x}_2$ and $\widehat{\zeta}_1$. When primary helium pressure is adopted as system output, $\widehat{\zeta}_2$ acts as a favorable indicator for helium-leakage detection.

Based on the above discussions about the simulation results for Cases A, B and C, $\widehat{\zeta}_2$ and $\widehat{\zeta}_1$ are suitable to be employed as indicators for reactivity-step and helium-flow-step when normalized neutron flux is adopted as the system output. While $\widehat{\zeta}_2$ is applicable for detecting helium leakages with primary helium pressure as system output. Table 2 summarizes pairing relationships among anomalies, system output and detection-oriented observation variables, laying foundations for anomaly detection system design. Detection thresholds can be determined following signal detection theory such as the Neyman-Pearson criterion, which maximizes detection accuracy while constraining false alarm rate. It should be highlighted that HoNN-ESO mainly contributes by generating high-quality state- and disturbance-observation signals as smarter detection features. Threshold tuning, false-alarm analysis and detection performance metrics lie outside the scope of this paper. As illustrated in Figs. 4–9, observations $\widehat{x}_2$, $\widehat{\zeta}_1$ and $\widehat{\zeta}_2$ respond to anomalies considerably faster than raw measured output, providing the capability of HoNN-ESO to supply advanced detection indicators.

Table 3
Single-step computation times of both ESO and HoNN-ESO with 10 and 40 hidden-layer neurons.

Observer	Hidden layer neuron number	Averaged single-step computation time (s)	Standard deviation (s)
ESO	0	1.0067×10^{-4}	0.9507×10^{-4}
HoNN-ESO	10	1.5753×10^{-4}	0.7128×10^{-4}
	40	5.6761×10^{-4}	1.0981×10^{-4}

To further validate the feasibility of applying the HoNN-ESO for generating detection-indicators, its robustness against measurement noises must be tested. As shown in Figs. 10 and 11, simulation results for measurement-noise tests reveal superior noise-resistant performance for HoNN-ESO compared with classical ESO (75), which stems from disturbance-compensation effects of the embedded MLP. Larger hidden-layer neuron number l strengthens learning ability and achieve better noise attenuation. According to Eq. (73), a larger positive constant a endows the eigenvalues of system matrix $\boldsymbol{A}$ defined by (74) with more negative real parts, which is beneficial for improving observation performance. From Figs. 10 and 11, increasing constant a from 0.68 to 0.8 assigns mitigates the overshoot of observation error $\widetilde{y}$, and oscillations can be further suppressed by raising neuron count from 10 to 40. As also evident from the responses of $\widehat{x}_2$, $\widehat{\zeta}_1$ and $\widehat{\zeta}_2$ in Fig. 11, steady-state fluctuations excited by measurement noise are smaller for HoNN-ESO, especially under higher neuron number l, confirming acceptable noise robustness.

For constant measurement bias conditions mimicking sensor faults, it can be seen from Fig. 12 that HoNN-ESO (72) converges toward new steady-state values faster than classical ESO (75). The embedded MLP rapidly compensates nonlinear dynamics triggered by large measurement bias, yielding smaller overshoots and shorter transition durations for $\widetilde{y}$, $\widehat{x}_2$, $\widehat{\zeta}_1$ and $\widehat{\zeta}_2$. Increasing neuron count l from 10 to 40 further reduces overshoots and shortens transients, demonstrating satisfactory bias-rejection robustness.

Computational overhead is assessed to verify real-time implementability. From (72) and (75), both observers are formulated as systems of ordinary differential equations (ODEs) that are solved online in a step-by-step manner, where the step size corresponds to the length of sampling period. The computation times of ESO and HoNN-ESO essentially determine whether these observers can be deployed online in real-time applications. For nuclear plant monitoring, sampling periods are typically no $<$0.1 s, and single-step computation time must be shorter than 0.1 s to guarantee real-time performance. Timing tests are completed within MATLAB/Simulink on hardware equipped with an Intel Ultra 9 285H CPU and an NVIDIA GeForce RTX 5060 GPU, adopting fixed-step Runge-Kutta solver with step size 0.1 s. Averaged single-step computation time and corresponding standard deviation are listed in Table 3. Here, the averaged single-step computation time is defined as the arithmetic mean value of the wall-clock time required to complete a single integration step, while the standard deviation of single-step computation time quantifies the variability of the single-step computation time across all steps in a simulation, computed as the square root of the average squared deviation from the mean. Although computation time increases rises with neuron number, all values remain on the order of 10^{-4}s, far below the 0.1 s sampling period requirement. This confirms real-time feasibility for both ESO and HoNN-ESO.

In summary, the MLP embedded within HoNN-ESO alleviates high-gain requirements for observers, and observation performance can be enhanced by increasing hidden-layer-neuron quantities. Simulation cases covering reactivity-step disturbances, helium-flowrate-step disturbances, helium-leakage-rate-step disturbances, measurement noise and measurement bias demonstrate that disturbance- and disturbance-derivative estimates serve as more sensitive anomaly-detection indicators than raw system outputs. Formulated as ordinary-differential-equation systems, HoNN-ESO is convenient for practical engineering

deployment.

5. Conclusions

External disturbances including modeling uncertainties constitute major sources of performance degradation and potential system anomalies. Disturbance observation is therefore critical for anomaly detection and disturbance attenuation. A core challenge for disturbance observer design arises from the lack of accurate system dynamic models. Conventional disturbance observers achieve compensation by boosting observer gains; nevertheless, high gains amplify measurement-noise sensitivity and complicate practical implementation. To remove reliance on high observer gains, this work embeds a single-hidden-layer MLP into the ESO framework and proposes the HoNN-ESO. Through appropriate configuration of MLP output gains, the KYP condition is no longer required. Theoretical analysis proves that HoNN-ESO delivers globally asymptotically bounded estimates for state variables, lumped disturbances and their high-order time derivatives over a defined set under given conditions. HoNN-ESO is applied to monitoring for modular high-temperature gas-cooled reactors. Numerical simulations under reactivity-step, primary-flowrate-step, helium-leakage-rate-step, measurement-noise and measurement-bias scenarios validate theoretical results and confirm that appropriately increasing hidden-layer-neuron number improves observation performance. Future research will focus on deploying HoNN-ESO onto digital-control-system platforms for monitoring advanced nuclear-power plants such as the under-construction HTR-PM600.

CRediT authorship contribution statement

Zhe Dong: Writing – original draft, Methodology, Investigation, Conceptualization. **Zijian Wu:** Writing – review & editing, Visualization, Validation, Software. **Xiaojin Huang:** Writing – review & editing, Supervision. **Lei Shi:** Project administration, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

Abdollahi, F., Talebi, H. A., & Patel, R. V. (2006). A stable neural network-based observer with application to flexible-joint manipulators. *IEEE Transactions on Neural Networks, 17*, 118–129.

Alanis, A. Y., Sanchez, E. N., Loukianov, A. G., & Perez, M. A. (2011). Real-time recurrent neural state-estimation. *IEEE Transactions on Neural Network, 22*, 497–505.

Cao, Y., Subhashri, A R, Chandrasekar, A., Radhika, T., & Przybyszewski, K. (2024). Exponential state estimation for delayed competitive neural network via stochastic sampled-data control with Markov jump parameters under actuator failure. *Journal of Artificial Intelligence and Soft Computing Research, 14*, 373–385.

Chairez, I. (2009). Wavelet differential neural network observer. *IEEE Transactions on Neural Networks, 20*, 1439–1449.

Chairez, I., Poznyak, A., & Poznyak, T. (2006). New sliding-mode learning law for dynamic neural network observer. *IEEE Transactions on Circuits and Systems—II: Express Briefs, 53*, 1338–1342.

Chairez, I., Poznyak, A., & Poznyak, T. (2009). Stable weights dynamics for a class of differential neural network observer. *IET Control Theory and Applications, 3*, 1437–1447.

Dinh, H. T., Kamalapurkar, R., Bhasin, S., & Dixon, W. E. (2014). Dynamic neural network-based robust observers for uncertain nonlinear systems. *Neural Networks, 60*, 44–52.

Dixon, W. E. (2023). Deep recurrent neural network-based observer for uncertain nonlinear systems. In *Proceedings of the 22nd IFAC World Congress, Yokohama, Japan*.

Dong, Z. (2022). Dynamical modeling and coordinated control design of a multimodular nuclear power-hydrogen cogeneration plant. *Energy Conversion and Management, 272*, Article 116369.

Dong, Z., Pan, Y., Dong, Z., Dong, Y., & Huang, X. (2018). Dynamical modeling and simulation of the six-modular high temperature gas-cooled reactor plant HTR-PM600. *Energy, 155*, 971–991.

Dong, Z., Zhang, Z., Dong, Y., Shi, L., Huang, X., Zhu, Y., & Jiang, D. (2025). Testing the feasibility of multi-modular design in an HTR-PM nuclear plant. *Nature Communications, 16*, 2778.

Griffis, E. J., Patil, O. S., Hart, R. G., & Dixon, W. E. (2024). Lyapunov-based long short-term memory (Lb-LSTM) neural network-based adaptive observer. *IEEE Control Systems Letters, 8*, 97–102.

Guo, B.-Z., & Zhao, Z.-L. (2011). On the convergence of an extended state observer for nonlinear systems with uncertainty. *Systems & Control Letters, 60*, 420–430.

Han, J. (2009). From PID to active disturbance rejection control. *IEEE Transactions on Industrial Electronics, 56*, 900–906.

Haykin, S. (2021). *Kalman filtering and neural networks*. New York, U.S.: Wiley.

Hochreiter, S., & Schmidhuber, J. (1997). Long short-term memory. *Neural Computation, 9*, 1735–1780.

Hu, J., Wu, W., Ji, B., & Wang, C. (2022). Observer design for sampled-data systems via deterministic learning. *IEEE Transactions on Neural Network and Learning Systems, 33*, 2931–2939.

Kim, Y. H., Lewis, F. L., & Abdallah, C. T. (1996). Nonlinear observer design using dynamic recurrent neural network. In *Proceedings of the 35th Conference on Decision and Control, Kobe, Japan*.

Kosmatopoulos, E. B., Polycarpou, M. M., Christodoulou, M. A., & Ioannou, P. A. (1995). High-order neural network structures for identification of dynamical systems. *IEEE Transactions on Neural Networks, 6*, 422–431.

LeCun, Y., Bengio, Y., & Hinton, G. (2015). Deep learning. *Nature, 521*, 436–444.

Li, J., & Dong, Z. (2025). Passivity-based control for methane steam reforming in nuclear cogeneration system. *International Journal of Hydrogen Energy, 140*, 473–482.

Li, J., He, D., Wang, X., & Kang, Y. (2025). BP neural network-based explicit MPC of nonlinear boiler-turbine systems. *Journal of Process Control, 145*, Article 103353.

Luenberger, D. G. (1964). Observing the state of a linear system. *IEEE Transactions on Military Electronics, 8*, 74–80.

Merabet, A., Kanukollu, S., Al-Durra, A., & El-Saadany, E. F. (2023). Adaptive recurrent neural network for uncertainties estimation in feedback control system. *Journal of Automation and Intelligence, 2*, 119–129.

Mhaskar, H. N. (1993). Approximation properties of a multilayer feedforward artificial neural network. *Advances in Computational Mathematics, 1*, 61–80.

Narendra, K. S., & Parthasarathy, K. (1990). Identification and control of dynamical systems using neural networks. *IEEE Transactions on Neural Networks, 1*, 4–27.

Pinkus, A. (1999). Approximation theory of the MLP model in neural networks. *Acta Numerica, 8*, 143–195.

Poznyak, A. S., Yu, W., Sanchez, E. N., & Perez, J. P. (1999). Nonlinear adaptive trajectory tracking using dynamic neural networks. *IEEE Transactions on Neural Networks, 10*, 1402–1411.

Rantzer, A. (1996). On the Kalman-Yakubovich-Popov lemma. *Systems & Control Letters, 28*, 7–10.

Ricalde, L. J., & Sanchez, E. N. (2005). Inverse optimal nonlinear recurrent high order neural observer. In *Proceedings of International Joint Conference on Neural Networks, Montreal, Canada*.

Rubio, J. D. J., Orozco, E., Cordova, D A, Hernandez, M A, Rosas, F J, & Pacheco, J. (2024). Observer-based differential evolution constrained control for safe reference tracking in robots. *Neural Networks, 175*, Article 106273.

Sanchez, E. N., & Ricalde, L. J. (2003). Trajectory tracking via adaptive recurrent neural control with input saturation. In *Proceedings of the International Joint Conference on Neural Networks, Portland, U.S.*.

Stepanyan, V., & Hovakimyan, N. (2007). Robust adaptive observer design for uncertain systems with bounded disturbances. *IEEE Transactions on Neural Networks, 18*, 1392–1403.

Subhashri, A R, Radhika, T., & Chandrasekar, A. (2026). Robust dissipative sliding mode control synchronization of memristive inertial competitive neural networks with time-varying delay. *The European Physical Journal Special Topics, 234*, 8403–8425. https://doi.org/10.1140/epjs/s11734-025-01653-5

Talebi, H. A., Khorasani, K., & Tafazoli, S. (2009). A recurrent neural-network-based sensor and actuator fault detection and isolation for nonlinear systems with application to the satellite's attitude control subsystem. *IEEE Transactions on Neural Networks, 20*, 45–60.

Vargas, J. A. R., & Hemerly, E. M. (2001). Adaptive observer for unknown general nonlinear systems. *IEEE Transactions on Systems, Man, and Cybernetics—Part B: Cybernetics, 31*, 683–690.

Wang, C., & Hill, D. J. (2010). Deterministic learning and nonlinear observer design. *Asian Journal of Control, 12*, 714–724.

Wang, T., & Li, Y. (2022). Neural-network adaptive output-feedback saturation control for uncertain active suspension systems. *IEEE Transactions on Cybernetics, 52*, 2022.

Xian, B., de Queiroz, M. S., Dawson, D. M., & McIntyre, M. L. (2004). A discontinuous output feedback controller and velocity observer for nonlinear mechanical systems. *Automatica, 40*, 695–700.

Zhang, Y., & Wang, F. (2022). Observer-based fixed-time neural control for a class of nonlinear system. *IEEE Transactions on Neural Networks and Learning Systems, 33*, 2892–2902.

Zhang, Z., Dong, Y., Li, F., Huang, X., Zheng, Y., Dong, Z., Zhang, H., Chen, Z., & Li, X. (2024). Loss of cooling tests to verify inherent safety feature in the world's first HTR-PM nuclear power plant. *Joule, 8*, 2146–2159.

Zhang, Z., Dong, Y., Li, F., Zhang, Z., Wang, H., Huang, X., Li, H., Liu, B., Wu, X., Wang, H., Diao, X., Zhang, H., & Wang, J. (2016). The Shandong Shidao bay 200 MWe high temperature gas-cooled reactor pebble-bed module (HTR-PM) demonstration power plant: An engineering and technological innovation. *Engineering, 2*, 112–118.

Zhang, Z., Dong, Y., Shi, Q., Li, F., & Wang, H. (2022). 600-MWe high temperature gas-cooled reactor nuclear power plant HTR-PM600. *Nuclear Science and Technique, 33*, 101.

Zhu, R., Chai, T., & Shao, C. (1997). Robust nonlinear adaptive observer design using dynamic recurrent neural networks. In *Proceedings of the American Control Conference, Albuquerque, NM, USA*.

Zhu, Y., Dong, Z., Cheng, Z., Huang, X., Dong, Y., & Zhang, Z. (2023). Neural network extended state-observer for energy system monitoring. *Energy, 263*, Article 125736.